

SparseStack Is an Optimal Oblivious Subspace Embedding

Diar Heidary

diyar.heydari83@sharif.edu*

Version 2.0 — September 2026

AI usage disclosure. The mathematical arguments presented here were derived by large language models under the author’s direction, specifically ChatGPT 5.6 Sol and Claude Fable 5. This document was produced and revised in collaboration with ChatGPT 5.6 Sol and Claude Fable 5.1. The main theorem is formally verified in Lean 4. A comparison with this version is recorded in Appendix A, and the author believes that the formal statements accurately represent the claims made here. The author takes full responsibility for the mathematical claims, the manuscript–formalization correspondence, all citations and attributions, and any remaining errors. For the source, Lean artifact, version history, and further information, visit github.com/DiarHaidary/nelson-nguyen-sparse-fock. The formal artifact is registered in the Palomar registry: palomar-registry.org/entry?id=PALOMAR-2026-08-23-000004&version=1.

Abstract

We prove that fully independent SparseStack achieves the oblivious subspace embedding parameters conjectured by Nelson and Nguyen (FOCS 2013): $m = O((d + \log(1/\delta))/\varepsilon^2)$ rows and $s = O(\log(d/\delta)/\varepsilon)$ nonzero entries per column for distortion ε and failure probability δ on any fixed d -dimensional subspace, with explicit constants. The proof turns random-matrix concentration into a problem in finite-dimensional linear algebra. A coupling first reduces the moment estimates to a model with independent finite-valued entries. We represent these variables by multiplication operators, so their matrix moments become exact matrix elements of a deterministic operator on a finite tensor product. The central estimate bounds the contribution of $\ell \geq 1$ occupied sites sharing the external factor $\mathbb{R}^d$ by $d + \ell - 1$ rather than $d\ell$, yielding additive dependence on the dimension and the moment order. This approach controls both spectral edges without Gaussian comparison. The main theorem has been formally verified in Lean 4.

*Unaffiliated work. This manuscript is the result of independent research and was not conducted as part of a research program at Sharif University of Technology, a master’s thesis, or a doctoral dissertation.

Contents

1	Introduction	3
1.1	Sparse subspace embeddings	3
1.2	Main result	3
1.3	Related work	4
1.4	Overview of the proof	5
1.5	Notation and conventions	6
2	Reduction to independent entries	6
2.1	The hollow Gram error	6
2.2	A coupling with an independent-entry vector	7
2.3	Convex-order transfer	7
3	A finite tensor-product model for the moments	8
3.1	Independent three-point entries	8
3.2	Multiplication by one entry	8
3.3	Sites, patterns, and the product space	9
3.4	The moment identity	10
3.5	Grading and the three bands	11
3.6	Band inventory	12
3.7	Row operators and a factorization identity	12
4	The shared-factor light-sector inequality	13
5	Row-local hop estimates	16
6	Band estimates	18
6.1	A block Cauchy–Schwarz inequality	18
6.2	The four Gram budgets	19
6.3	Heavy–heavy terms	19
6.4	Light–light terms	19
6.5	Mixed terms	20
6.6	The uniform band envelope	20
7	Proof of the main theorem	21
7.1	The moment bound	21
7.2	From moments to probability	21
7.3	Integer rounding and explicit constants	22
8	Scope and limitations	22
A	Verification status	23
B	Notation index	25
	References	25

1 Introduction

1.1 Sparse subspace embeddings

Let $U \in \mathbb{R}^{n \times d}$ have orthonormal columns, and write its rows as $u_i^\top$ with $u_i \in \mathbb{R}^d$, so that

$$\sum_{i=1}^n u_i u_i^\top = I_d. \quad (1)$$

A random matrix $\Pi \in \mathbb{R}^{m \times n}$ is an *oblivious subspace embedding* for d -dimensional subspaces with distortion ε and failure probability δ if, for every such U ,

$$\mathbb{P}\{\|U^\top \Pi^\top \Pi U - I_d\| > \varepsilon\} \leq \delta. \quad (2)$$

Since $x^\top (U^\top \Pi^\top \Pi U - I_d)x = \|\Pi U x\|^2 - \|x\|^2$, the complement of the event in (2) is exactly the event that $(1 - \varepsilon)\|x\|^2 \leq \|\Pi U x\|^2 \leq (1 + \varepsilon)\|x\|^2$ for all $x \in \mathbb{R}^d$; a bound on the Gram error controls both edges of the spectrum of ΠU .

This paper concerns the following sparse construction. Fix integers $s, b \geq 1$ and let $H_1, \dots, H_s \in \mathbb{R}^{b \times n}$ be independent CountSketch matrices [9]. That is, for each block $g \in [s]$ and each column $i \in [n]$, draw a row $h_g(i)$ uniformly from $[b]$ and a Rademacher sign σ_{gi} , all $2sn$ variables being mutually independent, and set

$$(H_g)_{ai} = \sigma_{gi} \mathbf{1}_{\{h_g(i)=a\}} \quad (a \in [b], i \in [n]). \quad (3)$$

Each column of each block has exactly one nonzero entry. The *fully independent SparseStack* matrix is

$$\Pi = s^{-1/2} \begin{bmatrix} H_1 \\ \vdots \\ H_s \end{bmatrix} \in \mathbb{R}^{m \times n}, \quad m = sb, \quad (4)$$

so every column of Π has exactly s nonzero entries, each equal to $\pm s^{-1/2}$. This is the block construction of Kane and Nelson [21] and one of the two OSNAP variants of Nelson and Nguyen [24]; the name SparseStack follows Camaño, Epperly, Meyer, and Tropp [7] and the problem list [2]. “Fully independent” refers to the mutual independence of all pairs $(h_g(i), \sigma_{gi})$; implementations with limited independence are not covered by our result.

The parameters $m = O((d + \log(1/\delta))/\varepsilon^2)$ and $s = O(\log(d/\delta)/\varepsilon)$ are the target proposed in [24]; the row count cannot be improved in general by the lower bound of [25], and the question for the SparseStack distribution is recorded as Problem 5.4 of [2].

1.2 Main result

Throughout, set

$$c_\star = (1 - e^{-1})^{-1} < 1.582, \quad C_{\text{band}} = 3 + \sqrt{2}, \quad \Lambda = 90c_\star^2 < 226. \quad (5)$$

Theorem 1.1 (Fully independent SparseStack). *Let $n, d \geq 1$, let $U \in \mathbb{R}^{n \times d}$ satisfy $U^\top U = I_d$, and let $0 < \varepsilon \leq 1$ and $0 < \delta \leq 1$. Define*

$$q = \max\left\{1, \left\lceil \log_2 \frac{d}{\delta} \right\rceil\right\}, \quad D_q = d + 2q + 1, \quad (6)$$

and choose

$$s = \left\lceil \frac{\Lambda(2q+1)}{\varepsilon} \right\rceil, \quad M_0 = \frac{\Lambda^2 D_q}{\varepsilon^2}, \quad b = \left\lceil \frac{M_0}{s} \right\rceil, \quad m = sb. \quad (7)$$

Then the SparseStack matrix (4) satisfies

$$\mathbb{P}\{\|U^\top \Pi^\top \Pi U - I_d\| > \varepsilon\} \leq \delta, \quad (8)$$

that is, with probability at least $1 - \delta$,

$$(1 - \varepsilon)\|x\|^2 \leq \|\Pi U x\|^2 \leq (1 + \varepsilon)\|x\|^2 \quad \text{for every } x \in \mathbb{R}^d. \quad (9)$$

The parameters satisfy

$$s < \frac{1356 q}{\varepsilon}, \quad m < \frac{306456 (d + q)}{\varepsilon^2}. \quad (10)$$

In particular $m = O((d + \log(d/\delta))/\varepsilon^2)$ and $s = O(\log(d/\delta)/\varepsilon)$. Since $\log d \leq d$ for $d \geq 1$, the row bound is equivalently $m = O((d + \log(1/\delta))/\varepsilon^2)$, as stated in the abstract. The numerical constants have not been optimized; see Section 8.

Remark 1.2 (Random input subspace). The theorem is proved for a fixed matrix U . If U is random, independent of Π , and satisfies $U^\top U = I_d$ almost surely, then conditioning on U and applying the theorem gives (8) unchanged, since the bound is uniform in U . A matrix U that depends on Π is not covered.

1.3 Related work

Sparse subspace embeddings. CountSketch, with a single nonzero per column, is an oblivious subspace embedding with $m = O(d^2/\varepsilon^2)$ rows [8]. Nelson and Nguyen [24] introduced OSNAP, with s nonzeros per column, proved that $m = O(d \log^8(d)/\varepsilon^2)$ and $s = O(\log^3(d)/\varepsilon)$ suffice, and conjectured that $m = O((d + \log(1/\delta))/\varepsilon^2)$ and $s = O(\log(d/\delta)/\varepsilon)$ suffice; the row count is optimal by their lower bound [25]. Cohen [13] obtained $m = O(d \log(d)/\varepsilon^2)$ with $s = O(\log(d)/\varepsilon)$ through trace inequalities. Chenakkod, Dereziński, Dong, and Rudelson [10] obtained the optimal dimension $m = O(d/\varepsilon^2)$ with $s = O(\log^4(d)/\varepsilon^6)$; Chenakkod, Dereziński, and Dong reduced the sparsity to $O(\log^2(d)/\varepsilon + \log^3(d))$ [11] and then, for $\varepsilon \geq d^{-O(1)}$, to $\tilde{O}(\log(d)/\varepsilon)$ with $m = \tilde{O}(d/\varepsilon^2)$, where $\tilde{O}$ hides sub-polylogarithmic factors in d [12]. These three works rely on the matrix universality and comparison results of Bandeira, Boedihardjo, and van Handel [3] and of Brailovskaya and van Handel [5]. Theorem 1.1 gives $m = O((d + \log(d/\delta))/\varepsilon^2)$ and $s = O(\log(d/\delta)/\varepsilon)$ for the fully independent SparseStack distribution, with explicit constants and without Gaussian comparison; it does not cover other OSNAP distributions or limited independence.

The lower edge and lower bounds on sparsity. Tropp [29, Theorem 3.6] uses Gaussian comparison to prove the lower-edge guarantee for the complex SparseStack construction obtained by stacking independent complex CountSketch blocks whose nonzero entries are Steinhaus phases. Writing s for Tropp's block count and m for his total row count, for lower distortion $\alpha \in (0, 1)$ and failure probability $p \in (0, 1]$, his result requires

$$s \geq 6\alpha^{-1} \log(d/p), \quad m \geq 16\alpha^{-2} (d \vee \log(d/p)),$$

and gives $\mathbb{P}\{\sigma_{\min}^2(\Phi U) > 1 - \alpha\} \geq 1 - p$ for every $U \in \mathbb{C}^{n \times d}$ with orthonormal columns. Thus it establishes the lower edge (injectivity), but it does not address the upper edge or the real Rademacher model used in this paper. Huang, Rudelson, and Tikhomirov [20] show that in the proportional regime $m = O(d)$ no sketch with $O(1)$ nonzeros per column can be an oblivious subspace injection at constant distortion, under a mild structural assumption; the logarithmic sparsity in Theorem 1.1 therefore cannot in general be replaced by a constant in that regime.

Methods. The operator model of Section 3 is the finite instance of a standard construction: multiplication by a finitely supported random variable is a Jacobi matrix in an orthogonal-polynomial basis, and the decomposition of that matrix into raising and lowering parts is the interacting Fock space of Accardi and Bożejko [1]. The use of such decompositions to compute spectral quantities by counting vacuum-to-vacuum paths is familiar from the quantum-decomposition method in spectral graph theory [17, 19]. Norm bounds for operator-valued sums in terms of their Gram operators, as in Section 6, are in the spirit of the noncommutative Khintchine inequalities [16, 6]. Bandeira, Cipolloni, Schröder, and van Handel [4, Section 4.1] obtain norm–moment comparisons for polynomials in free semicircular operators through ultracontractivity estimates. Our argument instead uses exact finite classical multiplication operators. Our E_{iid} is a degree-two matrix polynomial in independent three-point variables. An alternative general framework for polynomial random-matrix concentration is provided by Rajendran and Tulsiani [26], using recursive matrix Efron–Stein inequalities. Their Rademacher bounds reduce to deterministic coefficient matrices, and Section 8 treats sparse graph matrices with sparsity-dependent bounds. We have not checked whether their framework recovers the SparseStack parameters proved here.

The labelled shared-factor bound underlying Section 4 has an equivalent formulation as a maximally entangled-projector sum in quantum cloning [22, Section IV and Appendix A]. We give a self-contained proof and apply it through a hard-core compression in the band estimates. The reduction of Section 2 uses the conditional-expectation and convex-order comparison principle underlying Hoeffding’s sampling comparison [18] and its matrix counterpart due to Gross and Nemes [15]; the signed-selector coupling here compares different laws from their with- and without-replacement sampling models.

1.4 Overview of the proof

The proof bounds the even moments $\mathbb{E} \text{tr}(E^{2q})$ of the Gram error $E = U^\top \Pi^\top \Pi U - I_d$ and applies Markov’s inequality. It has four parts.

Reduction to independent entries (Section 2). Because every column of Π has unit norm, E is a hollow sum over pairs of distinct columns. A coupling replaces each signed one-hot column selector $\xi \in \mathbb{R}^b$ of a CountSketch block by a vector $\zeta \in \mathbb{R}^b$ with independent entries in $\{0, \pm 1\}$, in such a way that $\xi = \mathbb{E}[c_b \zeta \mid \xi]$ for an explicit constant $c_b \leq c_\star$. Conditional Jensen then gives $\mathbb{E} \text{tr}(E^{2q}) \leq c_b^{4q} \mathbb{E} \text{tr}(E_{\text{iid}}^{2q})$, where E_{iid} is the hollow Gram error of the independent-entry model.

A finite operator model (Section 3). The entries of the independent model take three values, so multiplication by an entry acts on a three-dimensional space as the Jacobi matrix (21). Tensoring over the mn entries, $\mathbb{E} \text{tr}(E_{\text{iid}}^k)$ becomes the vacuum matrix element of the k -th power of a deterministic operator $\mathcal{M}$ on $\mathbb{R}^d \otimes (\mathbb{R}^3)^{\otimes mn}$ (Proposition 3.1). The space is graded by the total level of the tensor factors, and $\mathcal{M}$ splits into three bands B^+, B^0, B^- that raise the grade by 2, preserve it, or lower it by 2.

Band estimates (Sections 4–6). The main technical result is the uniform bound

$$\|B^\Delta|_{\mathcal{K}_\nu}\| \leq (3 + \sqrt{2}) \left[\sqrt{\frac{d + \nu + 1}{m}} + \frac{d + \nu + 1}{m} + \frac{\nu + 1}{s} \right] \quad (11)$$

on the grade- ν sector $\mathcal{K}_\nu$ (Proposition 6.5). Each band is a sum of terms that act at two sites of one row of the sketch, and each such term factors through row operators built from the frame vectors u_i of that row; its norm is then bounded by the geometric mean of two Gram operators (Lemma 6.1). The square-root term in (11) comes from the creation of two light sites, and the essential point is that its bound involves $d + \nu$ rather than a product of d and ν . This is the content of Lemma 4.1: on configurations with ℓ light sites the relevant

Gram operator has norm at most $d + \ell - 1$. The mechanism is that all ℓ sites share one external factor $\mathbb{R}^d$. Each site contributes the rank-one operator $|\omega\rangle\langle\omega|$ for a vector ω of squared norm d , but two such operators attached to different sites overlap only through the orthogonal projection $UU^\top$, so the Gram matrix of the ℓ insertions has diagonal entries d and off-diagonal blocks of norm at most one. The term $(\nu + 1)/s$ comes from the heavy legs of the Jacobi matrix, whose coefficient $\sqrt{b - 1}$ is large but which act only at occupied sites; their norms are bounded by counting occupied sites in a row (Section 5), and the normalization $(b - 1)/m \leq 1/s$ turns the count into the sparsity scale. Mixed terms are handled by the same two ingredients and the inequality $2\sqrt{xy} \leq x + y$.

Moment expansion (Section 7). Expanding $\langle\Omega, \mathcal{M}^{2q}\Omega\rangle$ into 3^{2q} words in the three bands, each nonvanishing vacuum-to-vacuum word stays below grade $2q$, which gives $\mathbb{E} \operatorname{tr}(E_{\text{iid}}^{2q}) \leq d(3\beta_q)^{2q}$ with β_q the right side of (11) at $\nu = 2q$. With $m \gtrsim (d + q)/\varepsilon^2$ and $s \gtrsim q/\varepsilon$ the base $3c_b^2\beta_q/\varepsilon$ is below $1/2$, and $q = \max\{1, \lceil \log_2(d/\delta) \rceil\}$ makes the failure probability at most δ .

The theorem has also been formalized in Lean 4; see Appendix A.

1.5 Notation and conventions

We write $[k] = \{1, \dots, k\}$. All sums over $i \neq j$ are over ordered pairs. The norm $\|\cdot\|$ is the Euclidean norm of a vector or the operator norm of a linear map; $\langle\cdot, \cdot\rangle$ is the Euclidean inner product, extended to tensor products in the usual way. All spaces are real and finite-dimensional, tensor products may be read as Kronecker products, $\dagger$ denotes the adjoint (transpose), and $\preceq$ is the Loewner order. For vectors v, w we write $|v\rangle\langle w|$ for the rank-one map $x \mapsto \langle w, x \rangle v$. For an operator $\mathcal{T}$ and a subspace V of its domain, $\mathcal{T}|_V$ is the restriction to V and $\|\mathcal{T}|_V\|$ its norm. All probabilities and expectations refer to the sketching variables.

Throughout, $u_1, \dots, u_n$ are the rows of a fixed U with $U^\top U = I_d$, and $U_{ij} = u_i u_j^\top$. Thus $\|u_i\| \leq 1$, and for every $S \subset [n]$, every $x \in \mathbb{R}^d$, and all scalars $(a_i)_{i \in S}$,

$$\left\| \sum_{i \in S} u_i u_i^\top \right\| \leq 1, \quad \left\| \sum_{i \in S} a_i u_i \right\|^2 \leq \sum_{i \in S} a_i^2, \quad \sum_{i \in S} (u_i^\top x)^2 \leq \|x\|^2. \quad (12)$$

The last two inequalities are $\|U_S^\top\| \leq 1$ and $\|U_S\| \leq 1$ for the row submatrix U_S ; we call them the synthesis and analysis inequalities. Every application below involves distinct columns i of a single row r of the sketch.

2 Reduction to independent entries

2.1 The hollow Gram error

For $g \in [s]$ and $i \in [n]$, let $\xi_{gi} = \sigma_{gi} e_{h_g(i)} \in \mathbb{R}^b$ be the i -th column of H_g . Since $\|\xi_{gi}\| = 1$ and (1) holds,

$$U^\top \Pi^\top \Pi U = \frac{1}{s} \sum_{g=1}^s \sum_{i,j=1}^n \langle \xi_{gi}, \xi_{gj} \rangle u_i u_j^\top = I_d + \frac{1}{s} \sum_{g=1}^s \sum_{i \neq j} \langle \xi_{gi}, \xi_{gj} \rangle u_i u_j^\top.$$

Hence the Gram error is the hollow sum

$$E_{\text{SS}} := U^\top \Pi^\top \Pi U - I_d = \frac{1}{s} \sum_{g=1}^s \sum_{i \neq j} \langle \xi_{gi}, \xi_{gj} \rangle u_i u_j^\top. \quad (13)$$

The absence of diagonal terms is what makes the coupling below applicable.

2.2 A coupling with an independent-entry vector

Let $\zeta \in \mathbb{R}^b$ have independent coordinates with

$$\mathbb{P}\{\zeta_a = 1\} = \mathbb{P}\{\zeta_a = -1\} = \frac{1}{2b}, \quad \mathbb{P}\{\zeta_a = 0\} = 1 - \frac{1}{b} \quad (a \in [b]), \quad (14)$$

and put

$$p_0 = \mathbb{P}\{\zeta = 0\} = \left(1 - \frac{1}{b}\right)^b, \quad c_b = (1 - p_0)^{-1},$$

so that $c_1 = 1$. Given ζ , define a signed basis vector ξ as follows. If $\zeta \neq 0$, choose an index $\hat{a}$ uniformly from $\text{supp } \zeta = \{a : \zeta_a \neq 0\}$, independently of everything else, and set $\xi = \zeta_{\hat{a}} e_{\hat{a}}$. If $\zeta = 0$, choose $\hat{a}$ uniformly from $[b]$ and an independent Rademacher sign σ , and set $\xi = \sigma e_{\hat{a}}$.

Lemma 2.1 (Selector coupling). (i) ξ is uniformly distributed on the $2b$ signed basis vectors of $\mathbb{R}^b$ and is independent of $|\text{supp } \zeta|$.

(ii) $\mathbb{E}[\zeta \mid \xi] = c_b^{-1} \xi$; equivalently, $\xi = \mathbb{E}[c_b \zeta \mid \xi]$.

(iii) $1 \leq c_b \leq c_\star$.

Proof. (i) Fix $k \in [b]$ and $\sigma \in \{-1, 1\}$. Conditionally on $|\text{supp } \zeta| = k_0 \geq 1$, the support is a uniformly random k_0 -subset of $[b]$ and the signs on the support are independent Rademacher variables, so

$$\mathbb{P}\{\xi = \sigma e_k \mid |\text{supp } \zeta| = k_0\} = \frac{k_0}{b} \cdot \frac{1}{k_0} \cdot \frac{1}{2} = \frac{1}{2b},$$

the three factors being the probabilities that $k \in \text{supp } \zeta$, that $\hat{a} = k$ given $k \in \text{supp } \zeta$, and that $\zeta_k = \sigma$. The fallback rule gives the same value when $k_0 = 0$. Thus the conditional law of ξ given $|\text{supp } \zeta|$ is uniform on the $2b$ signed basis vectors, which is (i).

(ii) Fix $\xi = \sigma e_k$. On $\{\zeta \neq 0\}$ the construction selected the coordinate k , so $\zeta_k = \sigma$; on $\{\zeta = 0\}$, $\zeta_k = 0$. By (i), $\mathbb{P}\{\zeta \neq 0 \mid \xi = \sigma e_k\} = 1 - p_0$, and therefore $\mathbb{E}[\zeta_k \mid \xi = \sigma e_k] = (1 - p_0)\sigma$. For $a \neq k$, flipping the sign of ζ_a preserves the joint law of ζ and the auxiliary randomness, preserves $\text{supp } \zeta$ and the selected index, and hence preserves ξ ; so the conditional law of ζ_a given $\xi = \sigma e_k$ is symmetric and $\mathbb{E}[\zeta_a \mid \xi = \sigma e_k] = 0$. Together, $\mathbb{E}[\zeta \mid \xi] = (1 - p_0)\xi = c_b^{-1} \xi$.

(iii) $0 \leq p_0 \leq e^{-1}$, because $(1 - 1/b)^b \leq e^{-1}$. $\square$

2.3 Convex-order transfer

Take independent copies (ξ_{gi}, ζ_{gi}) of the coupled pair, one for each $(g, i) \in [s] \times [n]$. By Lemma 2.1(i) the vectors ξ_{gi} have the joint law of the columns of the blocks H_g , so we may realize E_{SS} through them. Define the independent-entry counterpart

$$E_{\text{iid}} = \frac{1}{s} \sum_{g=1}^s \sum_{i \neq j} \langle \zeta_{gi}, \zeta_{gj} \rangle u_i u_j^\top, \quad (15)$$

and let $\mathcal{G}_\xi$ be the σ -algebra generated by all ξ_{gi} . Conditionally on $\mathcal{G}_\xi$, the vectors ζ_{gi} are independent, and the conditional law of ζ_{gi} depends only on ξ_{gi} . Hence, for $i \neq j$, Lemma 2.1(ii) gives

$$\mathbb{E}[\langle \zeta_{gi}, \zeta_{gj} \rangle \mid \mathcal{G}_\xi] = \langle \mathbb{E}[\zeta_{gi} \mid \xi_{gi}], \mathbb{E}[\zeta_{gj} \mid \xi_{gj}] \rangle = c_b^{-2} \langle \xi_{gi}, \xi_{gj} \rangle,$$

and summing against the deterministic matrices $u_i u_j^\top$,

$$E_{\text{SS}} = \mathbb{E}[c_b^2 E_{\text{iid}} \mid \mathcal{G}_\xi]. \quad (16)$$

The restriction to $i \neq j$ is essential: by Lemma 2.1(i), $\mathbb{E}[\|\zeta_{gi}\|^2 \mid \xi_{gi}] = \mathbb{E}|\text{supp } \zeta_{gi}| = 1$, whereas $\|\xi_{gi}\|^2 = 1$, so a diagonal term would pick up the factor $c_b^2 \neq 1$ when $b > 1$.

Proposition 2.2 (Moment transfer). *For every integer $q \geq 1$,*

$$\mathbb{E} \operatorname{tr}(E_{\text{SS}}^{2q}) \leq c_b^{4q} \mathbb{E} \operatorname{tr}(E_{\text{iid}}^{2q}). \quad (17)$$

Proof. Both E_{SS} and E_{iid} are real symmetric, since the (j, i) term of each ordered sum is the transpose of the (i, j) term. On real symmetric matrices the function $f(A) = \operatorname{tr}(A^{2q}) = \|A\|_{S_{2q}}^{2q}$ is convex, being the $2q$ -th power of the Schatten norm. Conditional Jensen and (16) give $f(E_{\text{SS}}) = f(\mathbb{E}[c_b^2 E_{\text{iid}} \mid \mathcal{G}_\xi]) \leq \mathbb{E}[f(c_b^2 E_{\text{iid}}) \mid \mathcal{G}_\xi]$, and taking expectations yields $\mathbb{E}f(E_{\text{SS}}) \leq \mathbb{E}f(c_b^2 E_{\text{iid}}) = c_b^{4q} \mathbb{E}f(E_{\text{iid}})$. $\square$

Comparisons of this kind, in which a dependent model is replaced by an independent one through a conditional barycenter and Jensen’s inequality, go back to Hoeffding’s treatment of sampling without replacement [18]; the matrix version is due to Gross and Nesme [15] and is used, for example, in [28]. See [27] for the convex order in general. The specific ingredient here is the signed-selector coupling of Lemma 2.1, whose target law admits the representation of the next section; the sampling laws differ from those in the cited with- and without-replacement comparisons.

3 A finite tensor-product model for the moments

This section expresses $\mathbb{E} \operatorname{tr}(E_{\text{iid}}^k)$ as a matrix element of a deterministic operator on a finite tensor product. The construction is elementary linear algebra on a finite probability space; the words “site,” “level,” and “vacuum” name index sets and basis vectors and carry no further meaning.

3.1 Independent three-point entries

Identify the pair $(g, a) \in [s] \times [b]$ with the row $r = (g - 1)b + a \in [m]$ of Π , and set

$$\eta_{ri} = \sqrt{b} \zeta_{gi}(a) \quad ((r, i) \in [m] \times [n]).$$

The mn variables η_{ri} are independent, each with the law

$$\mathbb{P}\{\eta = \sqrt{b}\} = \mathbb{P}\{\eta = -\sqrt{b}\} = \frac{1}{2b}, \quad \mathbb{P}\{\eta = 0\} = 1 - \frac{1}{b}, \quad (18)$$

so that $\mathbb{E}\eta = 0$, $\mathbb{E}\eta^2 = 1$, and $\mathbb{E}\eta^4 = b$. Expanding the inner products in (15) and using $m = sb$,

$$E_{\text{iid}} = \frac{1}{m} \sum_{r=1}^m \sum_{i \neq j} \eta_{ri} \eta_{rj} u_i u_j^\top. \quad (19)$$

Under the parameter choice (7) one has $b \geq 2$ (see (90) below), so η takes each of its three values with positive probability. We assume $b \geq 2$ from now on.

3.2 Multiplication by one entry

Let $L_2(\eta)$ be the space of real functions on the three-point support of (18), with inner product $\langle f, g \rangle = \mathbb{E}[f(\eta)g(\eta)]$. It is three-dimensional, and the functions

$$e_0 = 1, \quad e_1 = \eta, \quad e_2 = \frac{\eta^2 - 1}{\sqrt{b - 1}} \quad (20)$$

form an orthonormal basis: the cross inner products vanish by centering and symmetry, $\mathbb{E}\eta^2 = 1$, and $\mathbb{E}(\eta^2 - 1)^2 = \mathbb{E}\eta^4 - 2\mathbb{E}\eta^2 + 1 = b - 1$. Since $\eta^3 = b\eta$ on the support, multiplication by η acts by

$$\eta e_0 = e_1, \quad \eta e_1 = \eta^2 = e_0 + \sqrt{b-1} e_2, \quad \eta e_2 = \frac{\eta^3 - \eta}{\sqrt{b-1}} = \sqrt{b-1} e_1,$$

so its matrix in the basis (e_0, e_1, e_2) is the Jacobi matrix

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & \sqrt{b-1} \\ 0 & \sqrt{b-1} & 0 \end{pmatrix}. \quad (21)$$

Write $|a\rangle\langle c|$ for the matrix unit that sends e_c to e_a and annihilates the other two basis vectors, and define the four *legs*

$$P^\dagger = |1\rangle\langle 0|, \quad P = |0\rangle\langle 1|, \quad R = |2\rangle\langle 1|, \quad R^\dagger = |1\rangle\langle 2|, \quad (22)$$

so that

$$J = P + P^\dagger + \sqrt{b-1} (R + R^\dagger). \quad (23)$$

The nonzero actions of the legs on the basis, and their coefficients in (23), are as follows; all other actions on basis vectors are zero.

Leg	Nonzero action	Change of index	Coefficient in J
$P^\dagger$	$e_0 \mapsto e_1$	+1	1
P	$e_1 \mapsto e_0$	-1	1
R	$e_1 \mapsto e_2$	+1	$\sqrt{b-1}$
$R^\dagger$	$e_2 \mapsto e_1$	-1	$\sqrt{b-1}$

Thus $P^\dagger$ and R raise the basis index and P and $R^\dagger$ lower it. We call $P, P^\dagger$ the *light* legs and $R, R^\dagger$ the *heavy* legs; the names refer to the coefficient $\sqrt{b-1}$ in (23), which is the only place where the sparsity enters the local model. The representation of a finitely supported random variable by a Jacobi matrix in an orthogonal-polynomial basis, and the decomposition of that matrix into raising and lowering parts (the diagonal part vanishes here because η is symmetric), is the finite case of the interacting Fock space construction of Accardi and Bożejko [1]; the three-level case used here is completely described by (21)–(23).

3.3 Sites, patterns, and the product space

Let $\mathbf{l} = [m] \times [n]$; its elements (r, i) are called *sites*. Let

$$\mathcal{F} = \bigotimes_{(r,i) \in \mathbf{l}} \mathbb{R}^3 = (\mathbb{R}^3)^{\otimes mn}, \quad (24)$$

with the factors ordered lexicographically, so that the site (r, i) occupies position $\text{pos}(r, i) = (r-1)n + i$. For a 3×3 matrix Z and a site (r, i) , the *lift* of Z to the site (r, i) is the $3^{mn} \times 3^{mn}$ matrix

$$Z_{ri} = I_3^{\otimes (\text{pos}(r,i)-1)} \otimes Z \otimes I_3^{\otimes (mn - \text{pos}(r,i))}, \quad (25)$$

which applies Z to the factor indexed by (r, i) and the identity to all other factors. Lifts to distinct sites commute, and $(Z_{ri})^\dagger = (Z^\dagger)_{ri}$. We write $J_{ri}, P_{ri}, P_{ri}^\dagger, R_{ri}, R_{ri}^\dagger$ for the lifts of the corresponding local matrices.

A *pattern* is a map $\tau : \mathbb{I} \rightarrow \{0, 1, 2\}$; the value τ_{ri} is the *level* of the site (r, i) . The pattern τ indexes the basis vector

$$|\tau\rangle = \bigotimes_{(r,i) \in \mathbb{I}} e_{\tau_{ri}} \in \mathcal{F}, \quad (26)$$

and these 3^{mn} vectors form an orthonormal basis of $\mathcal{F}$. The all-zero pattern gives the *vacuum* $\Omega = e_0^{\otimes mn}$. A site at level 0, 1, or 2 in τ is called *free*, *light*, or *heavy*, respectively, and we write

$$S_j(\tau) = \{(r, i) \in \mathbb{I} : \tau_{ri} = j\}, \quad S_j^r(\tau) = \{i \in [n] : \tau_{ri} = j\}, \quad F_r(\tau) = S_0^r(\tau) \quad (27)$$

for the set of level- j sites, the level- j columns of row r , and the free columns of row r . For a fixed row r , the pattern $\tau[a \mapsto j]$ agrees with τ except that the site (r, a) is assigned level j , and $\tau[a \mapsto j, c \mapsto k]$ is defined analogously for two sites of the same row. In this notation, the lift of a leg $Z = |a'\rangle\langle a|$ acts on basis vectors by

$$Z_{ri}|\tau\rangle = \begin{cases} |\tau[i \mapsto a']\rangle, & \tau_{ri} = a, \\ 0, & \text{otherwise,} \end{cases} \quad (28)$$

by (25) and (26).

The link with probability is the following. By independence, the products $\prod_{(r,i)} e_{\tau_{ri}}(\eta_{ri})$ form an orthonormal basis of the L_2 space of the array (η_{ri}) , so the map $\mathcal{U}$ defined by $\mathcal{U}|\tau\rangle = \prod_{(r,i)} e_{\tau_{ri}}(\eta_{ri})$ is an isometry from $\mathcal{F}$ onto that space. Under $\mathcal{U}$, the vacuum corresponds to the constant function 1 and J_{ri} to multiplication by η_{ri} . Consequently, for every polynomial p in the commuting operators J_{ri} ,

$$\langle \Omega, p((J_{ri}))\Omega \rangle = \mathbb{E} p((\eta_{ri})). \quad (29)$$

3.4 The moment identity

Let $q_1, \dots, q_d$ be the standard basis of $\mathbb{R}^d$, and define

$$\mathcal{M} = \frac{1}{m} \sum_{r=1}^m \sum_{i \neq j} U_{ij} \otimes J_{ri} J_{rj} \quad \text{on } \mathbb{R}^d \otimes \mathcal{F}. \quad (30)$$

We refer to the factor $\mathbb{R}^d$ as the *external factor*. The operator $\mathcal{M}$ is self-adjoint: $(U_{ij} \otimes J_{ri} J_{rj})^\dagger = U_{ji} \otimes J_{rj} J_{ri}$, the two local factors commute because $i \neq j$, and the ordered sum is invariant under $(i, j) \mapsto (j, i)$.

Proposition 3.1 (Vacuum-moment identity). *For every integer $k \geq 0$,*

$$\mathbb{E} \operatorname{tr}(E_{\text{iid}}^k) = \sum_{\alpha=1}^d \langle q_\alpha \otimes \Omega, \mathcal{M}^k(q_\alpha \otimes \Omega) \rangle. \quad (31)$$

Proof. Under the identification $I_d \otimes \mathcal{U}$, the operator $\mathcal{M}$ becomes multiplication of $\mathbb{R}^d$ -valued functions of the array (η_{ri}) by the matrix-valued function E_{iid} of (19), and $\mathcal{M}^k$ becomes multiplication by E_{iid}^k . The vector $q_\alpha \otimes \Omega$ corresponds to the constant function q_α , so the α -th summand in (31) equals $\mathbb{E}[q_\alpha^\top E_{\text{iid}}^k q_\alpha]$, and summing over α gives $\mathbb{E} \operatorname{tr}(E_{\text{iid}}^k)$.

The same identity can be verified term by term. Expanding the k -th power of (30),

$$\mathcal{M}^k = \frac{1}{m^k} \sum_{\substack{r_1, \dots, r_k \in [m] \\ i_t \neq j_t \ (1 \leq t \leq k)}} U_{i_1 j_1} \cdots U_{i_k j_k} \otimes \prod_{t=1}^k J_{r_t i_t} J_{r_t j_t},$$

and (29) gives

$$\langle q_\alpha \otimes \Omega, \mathcal{M}^k(q_\alpha \otimes \Omega) \rangle = \frac{1}{m^k} \sum_{\substack{r_1, \dots, r_k \\ i_t \neq j_t}} q_\alpha^\top U_{i_1 j_1} \cdots U_{i_k j_k} q_\alpha \mathbb{E} \prod_{t=1}^k \eta_{r_t i_t} \eta_{r_t j_t},$$

which is exactly the expansion of $\mathbb{E}[q_\alpha^\top E_{\text{iid}}^k q_\alpha]$ obtained by substituting (19) k times. $\square$

Example 3.2 (The case $k = 2$). Expanding (19) twice,

$$\mathbb{E} \text{tr}(E_{\text{iid}}^2) = \frac{1}{m^2} \sum_{r_1, r_2=1}^m \sum_{\substack{i_1 \neq j_1 \\ i_2 \neq j_2}} \mathbb{E}[\eta_{r_1 i_1} \eta_{r_1 j_1} \eta_{r_2 i_2} \eta_{r_2 j_2}] \text{tr}(U_{i_1 j_1} U_{i_2 j_2}).$$

Independence factors the expectation over distinct sites, and a site that occurs once contributes $\mathbb{E}\eta = 0$; since $i_t \neq j_t$, each factor $\eta_{r_t i_t} \eta_{r_t j_t}$ involves two distinct sites, so every site occurs twice exactly when $r_1 = r_2$ and $\{i_1, j_1\} = \{i_2, j_2\}$, and then the expectation is $\mathbb{E}\eta_{r_i}^2 \eta_{r_j}^2 = 1$. For each ordered pair (i_1, j_1) there are two admissible ordered pairs (i_2, j_2) , namely (i_1, j_1) and (j_1, i_1) , and the m choices of the common row cancel one factor $1/m$. Hence

$$\mathbb{E} \text{tr}(E_{\text{iid}}^2) = \frac{1}{m} \sum_{i \neq j} [\text{tr}(U_{ij} U_{ij}) + \text{tr}(U_{ij} U_{ji})] = \frac{1}{m} \sum_{i \neq j} [(u_i^\top u_j)^2 + \|u_i\|^2 \|u_j\|^2].$$

On the operator side, $J_{r_2 i_2} J_{r_2 j_2} \Omega$ is the basis vector with light sites (r_2, i_2) and (r_2, j_2) , since only $P^\dagger$ acts nontrivially on e_0 . The inner product of Ω with $J_{r_1 i_1} J_{r_1 j_1}$ applied to that vector is nonzero only if the second pair of legs returns both sites to level 0, which forces $r_1 = r_2$ and $\{i_1, j_1\} = \{i_2, j_2\}$; thus

$$\langle \Omega, J_{r_1 i_1} J_{r_1 j_1} J_{r_2 i_2} J_{r_2 j_2} \Omega \rangle = \mathbf{1}_{\{r_1=r_2, \{i_1, j_1\}=\{i_2, j_2\}\}},$$

and substituting into (31) reproduces the same sum. The vacuum matrix element is book-keeping for the surviving terms of the classical moment expansion.

3.5 Grading and the three bands

The *grade* of a pattern is

$$\text{gr}(\tau) = \sum_{(r,i) \in I} \tau_{ri} = |S_1(\tau)| + 2|S_2(\tau)|. \quad (32)$$

For an integer ν let

$$\mathcal{H}_\nu = \text{span}\{|\tau\rangle : \text{gr}(\tau) = \nu\} \subset \mathcal{F}, \quad \mathcal{K}_\nu = \mathbb{R}^d \otimes \mathcal{H}_\nu, \quad (33)$$

so that $\mathcal{F} = \bigoplus_{\nu=0}^{2mn} \mathcal{H}_\nu$ orthogonally, and let π_ν and $\mathcal{P}_\nu = I_d \otimes \pi_\nu$ be the orthogonal projections onto $\mathcal{H}_\nu$ and $\mathcal{K}_\nu$ (both zero if $\nu \notin \{0, \dots, 2mn\}$).

Each leg changes the level of one site by ± 1 and hence the grade by ± 1 . Expanding $J_{ri} J_{rj}$ by (23) expresses each summand of $\mathcal{M}$ as a sum of sixteen terms $c U_{ij} \otimes Z_{ri} W_{rj}$ with $Z, W \in \{P, P^\dagger, R, R^\dagger\}$ and $c \in \{1, \sqrt{b-1}, b-1\}$, each of which maps $\mathcal{K}_\nu$ into $\mathcal{K}_{\nu+2}$, $\mathcal{K}_\nu$, or $\mathcal{K}_{\nu-2}$. Consequently $\mathcal{P}_\mu \mathcal{M} \mathcal{P}_\nu = 0$ unless $\mu - \nu \in \{-2, 0, 2\}$, and

$$\mathcal{M} = B^+ + B^0 + B^-, \quad B^\Delta := \sum_\nu \mathcal{P}_{\nu+2\Delta} \mathcal{M} \mathcal{P}_\nu \quad (\Delta \in \{1, 0, -1\}), \quad (34)$$

where B^Δ maps $\mathcal{K}_\nu$ into $\mathcal{K}_{\nu+2\Delta}$. Since $\mathcal{M}$ is self-adjoint and the $\mathcal{P}_\nu$ are orthogonal projections,

$$(B^+)^\dagger = \sum_\nu (\mathcal{P}_{\nu+2} \mathcal{M} \mathcal{P}_\nu)^\dagger = \sum_\nu \mathcal{P}_\nu \mathcal{M} \mathcal{P}_{\nu+2} = \sum_\mu \mathcal{P}_{\mu-2} \mathcal{M} \mathcal{P}_\mu = B^-,$$

after the reindexing $\mu = \nu + 2$; similarly $(B^0)^\dagger = B^0$.

3.6 Band inventory

Put $h = b - 1$. In this subsection every sum is over $r \in [m]$ and ordered pairs $i \neq j$. By (28), a term $U_{ij} \otimes Z_{ri} W_{rj}$ with legs $Z = |a'\rangle\langle a|$ and $W = |c'\rangle\langle c|$ acts on $x \otimes |\tau\rangle$, $x \in \mathbb{R}^d$, by

$$(U_{ij} \otimes Z_{ri} W_{rj})(x \otimes |\tau\rangle) = \begin{cases} u_i (u_j^\top x) \otimes |\tau[i \mapsto a', j \mapsto c']\rangle, & \tau_{ri} = a \text{ and } \tau_{rj} = c, \\ 0, & \text{otherwise:} \end{cases} \quad (35)$$

the external vector is contracted against u_j and re-emitted along u_i , while the two sites change level. Define the light–light and heavy–heavy families

$$\begin{aligned} L_+ &= \sum U_{ij} \otimes P_{ri}^\dagger P_{rj}^\dagger, & L_0 &= \sum U_{ij} \otimes (P_{ri}^\dagger P_{rj} + P_{ri} P_{rj}^\dagger), & L_- &= L_+^\dagger, \\ H_+ &= \sum U_{ij} \otimes R_{ri} R_{rj}, & H_0 &= \sum U_{ij} \otimes (R_{ri} R_{rj}^\dagger + R_{ri}^\dagger R_{rj}), & H_- &= H_+^\dagger, \end{aligned} \quad (36)$$

and the mixed families

$$\begin{aligned} X_+ &= \sum U_{ij} \otimes P_{ri}^\dagger R_{rj}, & Y_+ &= \sum U_{ij} \otimes R_{ri} P_{rj}^\dagger, \\ X_0 &= \sum U_{ij} \otimes P_{ri}^\dagger R_{rj}^\dagger, & Y_0 &= \sum U_{ij} \otimes R_{ri}^\dagger P_{rj}^\dagger. \end{aligned} \quad (37)$$

For any two legs Z, W , using $U_{ij}^\dagger = U_{ji}$, commuting the factors at distinct sites, and exchanging the dummy indices i and j ,

$$\left(\sum U_{ij} \otimes Z_{ri} W_{rj} \right)^\dagger = \sum U_{ij} \otimes W_{ri}^\dagger Z_{rj}^\dagger; \quad (38)$$

thus, for instance, $X_+^\dagger = \sum U_{ij} \otimes R_{ri}^\dagger P_{rj}$ and $Y_0^\dagger = \sum U_{ij} \otimes P_{ri} R_{rj}$. Sorting the sixteen terms of the expansion of $J_{ri} J_{rj}$ by grade shift, and recording that a term with zero, one, or two heavy legs carries the coefficient 1, $\sqrt{h}$, or h , gives the decomposition

$$\begin{aligned} B^+ &= \frac{1}{m} [L_+ + \sqrt{h} (X_+ + Y_+) + h H_+], \\ B^0 &= \frac{1}{m} [L_0 + \sqrt{h} (X_0 + X_0^\dagger + Y_0 + Y_0^\dagger) + h H_0], \\ B^- &= \frac{1}{m} [L_- + \sqrt{h} (X_+^\dagger + Y_+^\dagger) + h H_-]. \end{aligned} \quad (39)$$

3.7 Row operators and a factorization identity

For a 3×3 matrix Z and a row r , define

$$\mathcal{S}_r(Z) = \sum_{i=1}^n u_i \otimes Z_{ri} : \mathcal{F} \rightarrow \mathbb{R}^d \otimes \mathcal{F}, \quad \mathcal{S}_r(Z)^\dagger = \sum_{i=1}^n u_i^\top \otimes (Z^\dagger)_{ri} : \mathbb{R}^d \otimes \mathcal{F} \rightarrow \mathcal{F}, \quad (40)$$

where u_i is regarded as the map $\mathbb{R} \rightarrow \mathbb{R}^d$, $t \mapsto t u_i$, and $u_i^\top$ as its adjoint. Thus $\mathcal{S}_r(Z)$ applies Z at one site of row r and synthesizes an external vector from the frame vectors of that row, while $\mathcal{S}_r(Z)^\dagger$ contracts the external vector against those frame vectors and applies $Z^\dagger$. The associated *Gram operator*

$$\mathcal{G}(Z) = \sum_{r=1}^m \mathcal{S}_r(Z) \mathcal{S}_r(Z)^\dagger = \sum_{r=1}^m \sum_{i,j=1}^n U_{ij} \otimes Z_{ri} (Z^\dagger)_{rj} \quad (41)$$

is positive semidefinite on $\mathbb{R}^d \otimes \mathcal{F}$. If Z is grade-homogeneous, in particular if Z is one of the four legs, then $\mathcal{S}_r(Z)$ and $\mathcal{S}_r(Z)^\dagger$ shift the grade by opposite amounts and $\mathcal{G}(Z)$ preserves grade.

Lemma 3.3 (Factorization identity). *For all 3×3 matrices Z, W ,*

$$\sum_{r=1}^m \sum_{i \neq j} U_{ij} \otimes Z_{ri} W_{rj} = \sum_{r=1}^m \mathcal{S}_r(Z) \mathcal{S}_r(W^\dagger)^\dagger - \sum_{r=1}^m \sum_{i=1}^n U_{ii} \otimes (ZW)_{ri}. \quad (42)$$

Proof. By (40), $\mathcal{S}_r(Z) \mathcal{S}_r(W^\dagger)^\dagger = \sum_{i,j} u_i u_j^\top \otimes Z_{ri} W_{rj}$, and the terms with $i = j$ are $U_{ii} \otimes (ZW)_{ri}$. $\square$

The local products that occur below are

$$\begin{aligned} (P^\dagger)^2 &= R^2 = P^\dagger R = P^\dagger R^\dagger = 0, \\ P^\dagger P &= R^\dagger R = |1\rangle\langle 1|, \quad PP^\dagger = |0\rangle\langle 0|, \quad RR^\dagger = |2\rangle\langle 2|. \end{aligned} \quad (43)$$

Define the diagonal operators

$$D_0 = \sum_{r,i} U_{ii} \otimes |0\rangle\langle 0|_{ri}, \quad D_1 = \sum_{r,i} U_{ii} \otimes |1\rangle\langle 1|_{ri}, \quad D_H = \sum_{r,i} U_{ii} \otimes (|1\rangle\langle 1| + |2\rangle\langle 2|)_{ri}, \quad (44)$$

the light Gram operator, and the transpose light hop

$$\hat{G} = \mathcal{G}(P^\dagger) = \sum_r \sum_{i,j} U_{ij} \otimes P_{ri}^\dagger P_{rj}, \quad H' = \sum_r \sum_{i \neq j} U_{ij} \otimes P_{ri} P_{rj}^\dagger. \quad (45)$$

By (38), H' is self-adjoint.

Corollary 3.4. *The following identities hold:*

$$L_+ = \sum_r \mathcal{S}_r(P^\dagger) \mathcal{S}_r(P)^\dagger, \quad H_+ = \sum_r \mathcal{S}_r(R) \mathcal{S}_r(R^\dagger)^\dagger, \quad (46)$$

$$X_+ = \sum_r \mathcal{S}_r(P^\dagger) \mathcal{S}_r(R^\dagger)^\dagger, \quad X_0 = \sum_r \mathcal{S}_r(P^\dagger) \mathcal{S}_r(R)^\dagger, \quad (47)$$

$$L_0 = (\hat{G} - D_1) + H', \quad \mathcal{G}(P) = D_0 + H', \quad (48)$$

$$H_0 = \mathcal{G}(R) + \mathcal{G}(R^\dagger) - D_H. \quad (49)$$

Proof. Apply Lemma 3.3 with (Z, W) equal to $(P^\dagger, P^\dagger)$, (R, R) , $(P^\dagger, R)$, and $(P^\dagger, R^\dagger)$; by (43) the diagonal term vanishes in each case, which gives (46) and (47). With $(Z, W) = (P^\dagger, P)$ the left side of (42) is the first half of L_0 and the right side is $\hat{G} - D_1$; adding H' gives the first identity in (48). With $(Z, W) = (P, P^\dagger)$ the left side is H' and the right side is $\mathcal{G}(P) - D_0$. Finally, $(Z, W) = (R, R^\dagger)$ and $(R^\dagger, R)$ express the two halves of H_0 as $\mathcal{G}(R) - \sum U_{ii} \otimes |2\rangle\langle 2|_{ri}$ and $\mathcal{G}(R^\dagger) - \sum U_{ii} \otimes |1\rangle\langle 1|_{ri}$. $\square$

The families Y_+ and Y_0 are not treated by factorization. For Y_+ the diagonal term in (42) would be $\sum U_{ii} \otimes (RP^\dagger)_{ri}$ with $RP^\dagger = |2\rangle\langle 0|$, a same-site transition from level 0 to level 2 that does not occur in $\mathcal{M}$; for Y_0 the diagonal term vanishes, but the resulting Cauchy–Schwarz bound would involve $\mathcal{G}(P)$, whose norm is of order m . Both families are bounded directly in Section 5, using the fact that a heavy leg can act only at an occupied site of the same row.

4 The shared-factor light-sector inequality

The operator $\hat{G} = \mathcal{G}(P^\dagger)$ moves one light site of a row to a free site of the same row, or leaves it in place through the terms $i = j$, with the external action U_{ij} . It preserves the set of heavy sites and the number of light sites. For a set $T \subset \mathbb{I}$ and an integer $\ell \geq 0$, write $|S; T\rangle$ for the basis vector with light sites S and heavy sites T , and let

$$F_T = \mathbb{I} \setminus T, \quad \mathbb{K}_{T,\ell} = \text{span}\{|S; T\rangle : S \subset F_T, |S| = \ell\}. \quad (50)$$

We call $K_{T,\ell}$ a *hard-core* block: its basis vectors are indexed by sets of ℓ distinct light sites, so no site is occupied twice, in contrast with the labelled tensor product introduced in the proof below, in which repetitions are allowed. Each $\mathbb{R}^d \otimes K_{T,\ell}$ is invariant under $\hat{G}$, and $\mathcal{H}_\nu = \bigoplus_{T: 2|T| \leq \nu} K_{T, \nu-2|T|}$.

Lemma 4.1 (Light-sector lemma). *On $\mathbb{R}^d \otimes K_{T,\ell}$, $\hat{G} = 0$ if $\ell = 0$, and*

$$0 \preceq \hat{G} \preceq (d + \ell - 1) I \quad \text{if } \ell \geq 1. \quad (51)$$

Consequently, for every $\kappa \geq 0$,

$$0 \preceq \hat{G} \preceq (d + \kappa - 1) I \preceq (d + \kappa) I \quad \text{on } \mathcal{K}_\kappa. \quad (52)$$

The point of the lemma is that the bound is additive in d and ℓ . Bounding the contribution of each of the ℓ light sites separately would give $d\ell$. The improvement comes from the fact that all light sites share a single external factor $\mathbb{R}^d$: each site contributes the rank-one operator $|\omega\rangle\langle\omega|$ for a vector ω of squared norm d , but two such operators attached to different sites overlap only through $UU^\top$, an operator of norm one.

Proof. If $\ell = 0$, every P_{rj} annihilates $K_{T,0}$ and $\hat{G} = 0$. Assume $\ell \geq 1$. The proof has three steps: we bound a sum of ℓ copies of a one-particle operator acting on different slots of a labelled tensor product, identify $\hat{G}$ as a compression of that sum, and conclude.

Step 1: a labelled model. Let $A = \mathbb{R}^d$, $R = \mathbb{R}^m$, $C = \mathbb{R}^n$, and $X = R \otimes C$; in this proof $e_r \in R$ and $e_i \in C$ denote standard basis vectors. For each row r put

$$w_r = \sum_{i=1}^n u_i \otimes e_r \otimes e_i \in A \otimes X, \quad Q = \sum_{r=1}^m |w_r\rangle\langle w_r|. \quad (53)$$

Since $\langle w_r, w_{r'} \rangle = \delta_{rr'} \sum_i \|u_i\|^2 = d \delta_{rr'}$, the vectors w_r are orthogonal with squared norm d ; up to a permutation of tensor factors, $Q = I_R \otimes |\omega\rangle\langle\omega|$ with $\omega = \sum_i u_i \otimes e_i \in A \otimes C$. On basis vectors,

$$Q(x \otimes e_r \otimes e_j) = \sum_{i=1}^n U_{ij} x \otimes e_r \otimes e_i \quad (x \in A), \quad (54)$$

because only w_r has a nonzero inner product with $x \otimes e_r \otimes e_j$, namely $\langle u_j, x \rangle$.

Let $X_1, \dots, X_\ell$ be labelled copies of X , with $X_k = R_k \otimes C_k$, and let $H_\ell = A \otimes X_1 \otimes \dots \otimes X_\ell$. Let Q_k be the operator Q acting on A and X_k , tensored with the identity on the other slots; all Q_k act on the same external factor A . We claim that

$$0 \preceq \sum_{k=1}^{\ell} Q_k \preceq (d + \ell - 1) I_{H_\ell}. \quad (55)$$

To prove this, let R_k^0 be a further copy of R , with basis vectors $e_r^{0,k}$, and set

$$W_k = R_k^0 \otimes \bigotimes_{j \neq k} X_j, \quad V_k(e_r^{0,k} \otimes \bigotimes_{j \neq k} x_j) = \sum_{i=1}^n u_i \otimes x_1 \otimes \dots \otimes (e_r \otimes e_i)_k \otimes \dots \otimes x_\ell, \quad (56)$$

which defines a linear map $V_k : W_k \rightarrow H_\ell$; the factors are placed in slot order. Thus V_k inserts w_r into the external factor and slot k and keeps the other slots. Its adjoint is

$$V_k^\dagger(x \otimes \bigotimes_{j=1}^{\ell} x_j) = \sum_{r=1}^m \sum_{i=1}^n \langle u_i, x \rangle \langle e_r \otimes e_i, x_k \rangle e_r^{0,k} \otimes \bigotimes_{j \neq k} x_j. \quad (57)$$

Composing (56) with (57) and comparing with (54) gives $V_k V_k^\dagger = Q_k$, and since the w_r are orthogonal with squared norm d ,

$$V_k^\dagger V_k = d I_{W_k}. \quad (58)$$

For $k \neq l$ we compute $V_k^\dagger V_l$ on a basis vector $e_{r'}^{0,l} \otimes (e_r \otimes e_j)_k \otimes \varphi$ of W_l , where φ collects the slots other than k and l . First V_l produces $\sum_i u_i \otimes (e_r \otimes e_j)_k \otimes (e_{r'} \otimes e_i)_l \otimes \varphi$; then $V_k^\dagger$ contracts the external factor against u_j and reads the row r from slot k , so that

$$V_k^\dagger V_l(e_{r'}^{0,l} \otimes (e_r \otimes e_j)_k \otimes \varphi) = e_r^{0,k} \otimes (e_{r'} \otimes \Gamma e_j)_l \otimes \varphi, \quad \Gamma = U U^\top. \quad (59)$$

Thus $V_k^\dagger V_l$ exchanges the two row registers and applies the orthogonal projection Γ to the column register of slot l ; in particular $\|V_k^\dagger V_l\| \leq 1$. Now let $V = [V_1 \cdots V_\ell] : \bigoplus_k W_k \rightarrow H_\ell$. Then $V V^\dagger = \sum_k V_k V_k^\dagger = \sum_k Q_k$, while $V^\dagger V$ is the block matrix with entries $V_k^\dagger V_l$, so that for $z = (z_1, \dots, z_\ell)$,

$$\langle z, V^\dagger V z \rangle \leq d \sum_k \|z_k\|^2 + \sum_{k \neq l} \|z_k\| \|z_l\| \leq (d + \ell - 1) \sum_k \|z_k\|^2,$$

using $\sum_{k \neq l} \|z_k\| \|z_l\| = (\sum_k \|z_k\|)^2 - \sum_k \|z_k\|^2 \leq (\ell - 1) \sum_k \|z_k\|^2$. Since $V V^\dagger$ and $V^\dagger V$ have the same nonzero eigenvalues, (55) follows.

Step 2: compression to the hard-core block. For a site $t = (r, i)$ write $e_t = e_r \otimes e_i \in X$. For $S = \{t_1, \dots, t_\ell\} \subset F_T$ define

$$\iota_T |S; T\rangle = \frac{1}{\sqrt{\ell!}} \sum_{\pi \in \mathfrak{S}_\ell} e_{t_{\pi(1)}} \otimes \cdots \otimes e_{t_{\pi(\ell)}} \in X_1 \otimes \cdots \otimes X_\ell. \quad (60)$$

The $\ell!$ tensor words are distinct and orthonormal, so ι_T is an isometry from $K_{T,\ell}$ into the symmetric tensors supported on distinct sites of F_T ; put $\mathcal{I}_T = I_d \otimes \iota_T$. Expanding (54), $Q_k = \sum_r \sum_{i,j} U_{ij} \otimes (|e_{(r,i)}\rangle \langle e_{(r,j)}|)_k \otimes I$. Fix $x \in \mathbb{R}^d$, a site $t = (r, j) \in S$, and a target $t' = (r, i)$ in the same row. Among the $\ell!$ words in (60) and the ℓ slots there are exactly $\ell!$ pairs (π, k) for which slot k carries t , and replacing t by t' in these words produces each word of the set $S' = (S \setminus \{t\}) \cup \{t'\}$ exactly once. If $t' \in T \cup (S \setminus \{t\})$, the resulting words are annihilated by $\iota_T^\dagger$, since they either involve a site of T or repeat a site. Otherwise the two normalizations $(\ell!)^{-1/2}$ combine with the count $\ell!$ to give coefficient one. Hence

$$\mathcal{I}_T^\dagger \left(\sum_{k=1}^\ell Q_k \right) \mathcal{I}_T (x \otimes |S; T\rangle) = \sum_{(r,j) \in S} \sum_{\substack{i \in [n] \\ (r,i) \notin T \cup (S \setminus \{(r,j)\})}} U_{ij} x \otimes |(S \setminus \{(r,j)\}) \cup \{(r,i)\}; T\rangle, \quad (61)$$

where the inner sum includes $i = j$. The right-hand side is exactly $\widehat{G}(x \otimes |S; T\rangle)$: P_{rj} removes the light site (r, j) , and $P_{ri}^\dagger$ recreates it at a site (r, i) that is free after the removal. Therefore

$$\widehat{G} = \mathcal{I}_T^\dagger \left(\sum_{k=1}^\ell Q_k \right) \mathcal{I}_T \quad \text{on } \mathbb{R}^d \otimes K_{T,\ell}. \quad (62)$$

Step 3: conclusion. Compressing (55) by the isometry $\mathcal{I}_T$ gives (51). For (52), decompose $\mathcal{K}_\kappa$ into the invariant blocks $\mathbb{R}^d \otimes K_{T,\ell}$ with $\ell = \kappa - 2|T| \leq \kappa$: on blocks with $\ell = 0$ the operator vanishes, and on the others $d + \ell - 1 \leq d + \kappa - 1$. Since $d \geq 1$, the constant $d + \kappa - 1$ is nonnegative in all cases. $\square$

Remark 4.2 (Relation to quantum cloning). The labelled bound (55) is equivalent to a known sharp spectral bound. Indeed, with $\Omega_d = \sum_{a=1}^d e_a \otimes e_a$,

$$\omega = \sum_i u_i \otimes e_i = (I_d \otimes U) \Omega_d.$$

After identifying each column-space factor $\text{im}(U)$ through U and omitting row-register identity factors, $\sum_{k=1}^\ell Q_k$ becomes $d \sum_{k=1}^\ell P_{0k}$, where P_{0k} projects onto $\Omega_d/\sqrt{d}$ on the common external factor and slot k , and acts as the identity on the other slots. The equal-weight specialization of Kay, Ramanathan, and Kaszlikowski [22, Section IV and Appendix A] gives $\|d \sum_{k=1}^\ell P_{0k}\| = d + \ell - 1$ for $\ell \geq 1$. Thus the labelled bound is attained and cannot be improved before compression. For $n > d$, decomposing each column slot into $\text{im}(U)$ and its orthogonal complement leaves the same sum with $j \leq \ell$ active slots, bounded by $d + j - 1$ when $j \geq 1$ and zero when $j = 0$. The proof above gives an elementary insertion-map argument and then compresses to the distinct-site sector; compression preserves the upper bound, but need not preserve equality.

5 Row-local hop estimates

This section bounds operators that act at one or two sites of a single row, with the external action determined by the frame vectors of that row. The estimates depend only on counting sites at a given level. For $\psi \in \mathbb{R}^d \otimes \mathcal{F}$ write $\psi = \sum_\tau x_\tau \otimes |\tau\rangle$ with $x_\tau \in \mathbb{R}^d$, so that $\|\psi\|^2 = \sum_\tau \|x_\tau\|^2$. For an operator $\mathcal{T}$ with values in $\mathbb{R}^d \otimes \mathcal{F}$ and a pattern σ , let $(\mathcal{T}\psi)_\sigma = (I_d \otimes \langle \sigma |) \mathcal{T} \psi \in \mathbb{R}^d$ denote the coefficient of $|\sigma\rangle$; for an operator with values in $\mathcal{F}$, $(\mathcal{T}\psi)_\sigma = \langle \sigma | \mathcal{T} \psi \rangle \in \mathbb{R}$. We use the row-subset inequalities (12) and the elementary inequality $\|\sum_{\alpha \in I} z_\alpha\|^2 \leq |I| \sum_{\alpha \in I} \|z_\alpha\|^2$. For a row r let

$$\nu_r(\tau) = |S_1^r(\tau)| + 2|S_2^r(\tau)|$$

be the grade of row r in the pattern τ , so that $\sum_r \nu_r(\tau) = \text{gr}(\tau)$, and let $\mathcal{N}_r$ be the operator on $\mathbb{R}^d \otimes \mathcal{F}$ with $\mathcal{N}_r(x \otimes |\tau\rangle) = \nu_r(\tau) x \otimes |\tau\rangle$.

In the following lemma, $Z = |a'\rangle\langle a|$ and $W = |c'\rangle\langle c|$ denote arbitrary legs, so that Z changes the level of a site from a to a' and W from c to c' ; we use the convention $\max \emptyset = 0$.

Lemma 5.1 (Row-local hop estimate). *Let $Z = |a'\rangle\langle a|$ and $W = |c'\rangle\langle c|$ be legs.*

- (i) (One site.) *Fix a row r and an integer $\ell \geq 0$, and let ψ be supported on patterns τ with $\nu_r(\tau) = \ell$. Then the one-site map $\mathcal{T}_r = \sum_i u_i^\top \otimes Z_{ri} = \mathcal{S}_r(Z^\dagger)^\dagger$ satisfies*

$$\|\mathcal{T}_r \psi\|^2 \leq N \|\psi\|^2, \quad N = \max\{|S_{a'}^r(\sigma)| : \nu_r(\sigma) = \ell + a' - a\}. \quad (63)$$

- (ii) (Two sites.) *Let $\mathcal{T} = \sum_r \sum_{i \neq j} U_{ij} \otimes Z_{ri} W_{rj}$. For every $\nu \geq 0$ and $\psi \in \mathcal{K}_\nu$,*

$$\|\mathcal{T} \psi\|^2 \leq N_{\text{out}} N_{\text{in}} \|\psi\|^2, \quad (64)$$

where $N_{\text{in}} = \max\{|S_a(\tau)| : \text{gr}(\tau) = \nu\}$ and $N_{\text{out}} = \max\{|S_{c'}(\sigma)| : \text{gr}(\sigma) = \nu + (a' - a) + (c' - c), S_{a'}(\sigma) \neq \emptyset\}$.

Proof. (i) For a pattern σ with $\nu_r(\sigma) = \ell + a' - a$, the coefficient of $|\sigma\rangle$ in $\mathcal{T}_r \psi$ collects the inputs that differ from σ at one site of row r where σ has level a' and the input has level a :

$$(\mathcal{T}_r \psi)_\sigma = \sum_{i \in S_{a'}^r(\sigma)} u_i^\top x_{\sigma[i \rightarrow a]},$$

with $x_\tau = 0$ when τ is outside the support of ψ . Hence

$$\begin{aligned}\|\mathcal{T}_r\psi\|^2 &= \sum_{\sigma} \left| \sum_{i \in S_{a'}^r(\sigma)} u_i^\top x_{\sigma[i \mapsto a]} \right|^2 \leq N \sum_{\sigma} \sum_{i \in S_{a'}^r(\sigma)} (u_i^\top x_{\sigma[i \mapsto a]})^2 \\ &= N \sum_{\tau} \sum_{i \in S_a^r(\tau)} (u_i^\top x_\tau)^2 \leq N \sum_{\tau} \|x_\tau\|^2,\end{aligned}$$

where the equality is the change of variables $(\sigma, i) \mapsto (\tau, i) = (\sigma[i \mapsto a], i)$, a bijection onto pairs with $i \in S_a^r(\tau)$ whose inverse is $\sigma = \tau[i \mapsto a']$, and the last step is the analysis inequality in (12).

(ii) For an output pattern σ , the coefficient $(\mathcal{T}\psi)_\sigma$ collects the inputs τ and ordered pairs (i, j) with $\tau_{ri} = a$, $\tau_{rj} = c$, and $\tau[i \mapsto a', j \mapsto c'] = \sigma$; given σ, r, i, j , the input is $\tau = \sigma[i \mapsto a, j \mapsto c]$. Hence

$$(\mathcal{T}\psi)_\sigma = \sum_r \sum_{j \in S_c^r(\sigma)} z_{\sigma, r, j}, \quad z_{\sigma, r, j} = \sum_{i \in S_a^r(\sigma) \setminus \{j\}} u_i (u_j^\top x_{\sigma[i \mapsto a, j \mapsto c]}).$$

If $S_{a'}(\sigma) = \emptyset$ the coefficient vanishes, so we may restrict to output patterns with $S_{a'}(\sigma) \neq \emptyset$, for which the number of pairs (r, j) in the outer sum is $|S_c^r(\sigma)| \leq N_{\text{out}}$. By the finite-family inequality, and then the synthesis inequality in (12) applied for fixed (σ, r, j) to the distinct columns i of row r ,

$$\|\mathcal{T}\psi\|^2 \leq N_{\text{out}} \sum_{\sigma} \sum_r \sum_{j \in S_c^r(\sigma)} \|z_{\sigma, r, j}\|^2 \leq N_{\text{out}} \sum_{\sigma} \sum_r \sum_{j \in S_c^r(\sigma)} \sum_{i \in S_a^r(\sigma) \setminus \{j\}} (u_j^\top x_{\sigma[i \mapsto a, j \mapsto c]})^2.$$

The map $(\sigma, r, i, j) \mapsto (\tau, r, i, j)$ with $\tau = \sigma[i \mapsto a, j \mapsto c]$ is a bijection from the index set of the last sum onto the set of (τ, r, i, j) with $\text{gr}(\tau) = \nu$, $i \in S_a^r(\tau)$, $j \in S_c^r(\tau)$, and $i \neq j$; its inverse is $\sigma = \tau[i \mapsto a', j \mapsto c']$. Re-indexing, and applying the analysis inequality in (12) to the sum over j for fixed (τ, r, i) ,

$$\|\mathcal{T}\psi\|^2 \leq N_{\text{out}} \sum_{\tau} \sum_r \sum_{i \in S_a^r(\tau)} \sum_{j \in S_c^r(\tau) \setminus \{i\}} (u_j^\top x_\tau)^2 \leq N_{\text{out}} \sum_{\tau} |S_a(\tau)| \|x_\tau\|^2 \leq N_{\text{out}} N_{\text{in}} \|\psi\|^2. \quad \square$$

Corollary 5.2 (Directional bounds). *For every $\nu \geq 0$,*

$$\|Y_+|_{\mathcal{K}_\nu}\| \leq \nu, \quad \|Y_0|_{\mathcal{K}_\nu}\| \leq \frac{\nu}{\sqrt{2}}, \quad \|H'|_{\mathcal{K}_\nu}\| \leq \nu. \quad (65)$$

Proof. Apply Lemma 5.1(ii). For Y_+ we have $(Z, W) = (R, P^\dagger)$, so $(a, a', c, c') = (1, 2, 0, 1)$: the outputs have grade $\nu + 2$ and at least one heavy site, hence $|S_1(\sigma)| = \nu + 2 - 2|S_2(\sigma)| \leq \nu$, while $|S_1(\tau)| \leq \nu$ for inputs; thus $\|Y_+\psi\|^2 \leq \nu^2 \|\psi\|^2$. For Y_0 we have $(Z, W) = (R^\dagger, P^\dagger)$ and $(a, a', c, c') = (2, 1, 0, 1)$: the outputs have grade ν , so $|S_1(\sigma)| \leq \nu$, while $|S_2(\tau)| \leq \nu/2$ for inputs; thus $\|Y_0\psi\|^2 \leq \nu^2 \|\psi\|^2/2$. For H' we have $(Z, W) = (P, P^\dagger)$ and $(a, a', c, c') = (1, 0, 0, 1)$: outputs have grade ν and $|S_1(\sigma)| \leq \nu$, and $|S_1(\tau)| \leq \nu$ for inputs. $\square$

In each case the counts that enter are the number of light sites of the output and the number of level- a sites of the input, both of order ν . Applied to L_+ , the same lemma would produce the number of free sites of the input, which is of order mn ; for this reason L_+ is treated by Lemma 4.1 instead. Table 1 summarizes the five applications of Lemma 5.1 used in this paper.

Corollary 5.3 (Heavy Gram budgets). *For every row r ,*

$$\mathcal{S}_r(R)\mathcal{S}_r(R)^\dagger \preceq \mathcal{N}_r \quad \text{and} \quad \mathcal{S}_r(R^\dagger)\mathcal{S}_r(R^\dagger)^\dagger \preceq \mathcal{N}_r.$$

Consequently,

$$\mathcal{G}(R) \preceq \nu I \quad \text{and} \quad \mathcal{G}(R^\dagger) \preceq \nu I \quad \text{on } \mathcal{K}_\nu. \quad (66)$$

Operator	(Z, W) or Z	(a, a')	(c, c')	Counts	Bound
Y_+	$(R, P^\dagger)$	$(1, 2)$	$(0, 1)$	$N_{\text{out}} \leq \nu, N_{\text{in}} \leq \nu$	$\ Y_+ _{\mathcal{K}_\nu}\ \leq \nu$
Y_0	$(R^\dagger, P^\dagger)$	$(2, 1)$	$(0, 1)$	$N_{\text{out}} \leq \nu, N_{\text{in}} \leq \nu/2$	$\ Y_0 _{\mathcal{K}_\nu}\ \leq \nu/\sqrt{2}$
H'	$(P, P^\dagger)$	$(1, 0)$	$(0, 1)$	$N_{\text{out}} \leq \nu, N_{\text{in}} \leq \nu$	$\ H' _{\mathcal{K}_\nu}\ \leq \nu$
$\mathcal{S}_r(R)^\dagger$	$R^\dagger$	$(2, 1)$	—	$N \leq (\ell - 1)_+$	$\mathcal{S}_r(R)\mathcal{S}_r(R)^\dagger \preceq \mathcal{N}_r$
$\mathcal{S}_r(R^\dagger)^\dagger$	R	$(1, 2)$	—	$N \leq \lfloor (\ell + 1)/2 \rfloor$	$\mathcal{S}_r(R^\dagger)\mathcal{S}_r(R^\dagger)^\dagger \preceq \mathcal{N}_r$

Table 1: The applications of Lemma 5.1. The first three rows use part (ii) on $\mathcal{K}_\nu$ (Corollary 5.2); the outputs have grade $\nu + 2$, ν , and ν , respectively. The last two rows use part (i) on the row-grade stratum ℓ (Corollary 5.3), where the image lies in the stratum $\ell - 1$, respectively $\ell + 1$; in both cases $N \leq \ell$ for $\ell \geq 1$, and the operator vanishes on the stratum $\ell = 0$.

Proof. Write $\psi = \sum_\ell \psi_{r,\ell}$ with $\psi_{r,\ell}$ supported on patterns with $\nu_r(\tau) = \ell$. The map $\mathcal{S}_r(R)^\dagger = \sum_i u_i^\top \otimes R_{ri}^\dagger$ is the one-site map of Lemma 5.1(i) with $Z = R^\dagger$, $(a, a') = (2, 1)$; it sends the stratum ℓ into the stratum $\ell - 1$, so the images of different strata are orthogonal, and

$$\begin{aligned} \langle \psi, \mathcal{S}_r(R)\mathcal{S}_r(R)^\dagger \psi \rangle &= \sum_{\ell \geq 1} \|\mathcal{S}_r(R)^\dagger \psi_{r,\ell}\|^2 \leq \sum_{\ell \geq 1} \max\{|S_1^r(\sigma)| : \nu_r(\sigma) = \ell - 1\} \|\psi_{r,\ell}\|^2 \\ &\leq \sum_{\ell \geq 1} \ell \|\psi_{r,\ell}\|^2 = \langle \psi, \mathcal{N}_r \psi \rangle, \end{aligned}$$

where the $\ell = 0$ term vanishes and, for $\ell \geq 1$, $|S_1^r(\sigma)| \leq \nu_r(\sigma) = \ell - 1 \leq \ell$. Similarly, $\mathcal{S}_r(R^\dagger)^\dagger = \sum_i u_i^\top \otimes R_{ri}$ has $(a, a') = (1, 2)$ and sends the stratum ℓ into the stratum $\ell + 1$; the relevant count is $|S_2^r(\sigma)| \leq \lfloor (\ell + 1)/2 \rfloor \leq \ell$ for $\ell \geq 1$, while for $\ell = 0$ the row has no light site and the image is zero. Summing over r and using $\sum_r \mathcal{N}_r = \nu I$ on $\mathcal{K}_\nu$ gives (66). $\square$

6 Band estimates

6.1 A block Cauchy–Schwarz inequality

Lemma 6.1. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be finite-dimensional Hilbert spaces, let $\mathcal{P}_{\text{in}}, \mathcal{P}_{\text{mid}}, \mathcal{P}_{\text{out}}$ be orthogonal projections on them, and let $A_r : \mathcal{X} \rightarrow \mathcal{Y}$ and $B_r : \mathcal{Y} \rightarrow \mathcal{Z}$ for $r \in [m]$. Then*

$$\left\| \mathcal{P}_{\text{out}} \left(\sum_r B_r \mathcal{P}_{\text{mid}} A_r \right) \mathcal{P}_{\text{in}} \right\|^2 \leq \left\| \mathcal{P}_{\text{in}} \left(\sum_r A_r^\dagger \mathcal{P}_{\text{mid}} A_r \right) \mathcal{P}_{\text{in}} \right\| \cdot \left\| \mathcal{P}_{\text{out}} \left(\sum_r B_r \mathcal{P}_{\text{mid}} B_r^\dagger \right) \mathcal{P}_{\text{out}} \right\|. \quad (67)$$

Proof. Let $\mathcal{A} : \text{ran } \mathcal{P}_{\text{in}} \rightarrow \bigoplus_r \text{ran } \mathcal{P}_{\text{mid}}$, $\mathcal{A}x = (\mathcal{P}_{\text{mid}} A_r x)_r$, and $\mathcal{B} : \bigoplus_r \text{ran } \mathcal{P}_{\text{mid}} \rightarrow \text{ran } \mathcal{P}_{\text{out}}$, $\mathcal{B}(y_r)_r = \sum_r \mathcal{P}_{\text{out}} B_r y_r$. Then $\mathcal{B}\mathcal{A}$ is the operator on the left of (67), while $\mathcal{A}^\dagger \mathcal{A}$ and $\mathcal{B}\mathcal{B}^\dagger$ are the two operators on the right, and $\|\mathcal{B}\mathcal{A}\|^2 \leq \|\mathcal{A}\|^2 \|\mathcal{B}\|^2 = \|\mathcal{A}^\dagger \mathcal{A}\| \|\mathcal{B}\mathcal{B}^\dagger\|$. $\square$

We apply the lemma to the factorized families of Corollary 3.4. For a leg Z let $\delta_Z \in \{1, -1\}$ be its grade shift, so that both $\mathcal{S}_r(Z)$ and $\mathcal{S}_r(Z^\dagger)^\dagger = \sum_i u_i^\top \otimes Z_{ri}$ shift the grade by δ_Z .

Corollary 6.2. *For legs Z, W and every $\nu \geq 0$,*

$$\left\| \left(\sum_r \mathcal{S}_r(Z) \mathcal{S}_r(W^\dagger)^\dagger \right) \Big|_{\mathcal{K}_\nu} \right\|^2 \leq \|\mathcal{G}(W^\dagger)|_{\mathcal{K}_\nu}\| \cdot \|\mathcal{G}(Z)|_{\mathcal{K}_{\nu+\delta_W+\delta_Z}}\|. \quad (68)$$

Proof. Apply Lemma 6.1 with $A_r = \mathcal{S}_r(W^\dagger)^\dagger$, $B_r = \mathcal{S}_r(Z)$, $\mathcal{P}_{\text{in}} = \mathcal{P}_\nu$, $\mathcal{P}_{\text{mid}} = \pi_{\nu+\delta_W}$, and $\mathcal{P}_{\text{out}} = \mathcal{P}_{\nu+\delta_W+\delta_Z}$. By grade homogeneity, $A_r \mathcal{P}_\nu = \pi_{\nu+\delta_W} A_r \mathcal{P}_\nu$ and $B_r \pi_{\nu+\delta_W} =$

$\mathcal{P}_{\nu+\delta_W+\delta_Z} B_r \pi_{\nu+\delta_W}$, so the left side of (67) is the left side of (68). On the right, since $\pi_{\nu+\delta_W} \preceq I$,

$$\sum_r A_r^\dagger \pi_{\nu+\delta_W} A_r \preceq \sum_r A_r^\dagger A_r = \mathcal{G}(W^\dagger), \quad \sum_r B_r \pi_{\nu+\delta_W} B_r^\dagger \preceq \sum_r B_r B_r^\dagger = \mathcal{G}(Z);$$

both Gram operators preserve grade, and compressing a Loewner inequality between positive operators to a grade sector preserves it. $\square$

6.2 The four Gram budgets

Proposition 6.3. *For every $\kappa \geq 0$, on $\mathcal{K}_\kappa$,*

$$\mathcal{G}(P^\dagger) = \widehat{G} \preceq (d + \kappa - 1) I, \quad \mathcal{G}(P) \preceq (m + \kappa) I, \quad \mathcal{G}(R) \preceq \kappa I, \quad \mathcal{G}(R^\dagger) \preceq \kappa I. \quad (69)$$

Moreover $0 \preceq D_0 \preceq mI$ on $\mathbb{R}^d \otimes \mathcal{F}$, while $0 \preceq D_1 \preceq \kappa I$ and $0 \preceq D_H \preceq \kappa I$ on $\mathcal{K}_\kappa$.

Proof. The first bound is Lemma 4.1, and the last two are Corollary 5.3. The diagonal operators of (44) are block diagonal in the pattern basis: on $x \otimes |\tau\rangle$ they act on the external vector by $\sum_r \sum_{i \in F_r(\tau)} u_i u_i^\top$, by $\sum_{(r,i) \in S_1(\tau)} u_i u_i^\top$, and by $\sum_{(r,i) \in S_1(\tau) \cup S_2(\tau)} u_i u_i^\top$, respectively. Each row contributes a matrix $\sum_{i \in S} u_i u_i^\top \preceq I_d$ by (12); there are m rows in the first case, while in the other two cases the number of contributing rows is at most the number of occupied sites, which is at most $\text{gr}(\tau) = \kappa$. Finally, $\mathcal{G}(P) = D_0 + H'$ by (48), and since H' is self-adjoint with $\|H'\|_{\mathcal{K}_\kappa} \leq \kappa$ by Corollary 5.2, $H' \preceq \kappa I$ on $\mathcal{K}_\kappa$. $\square$

6.3 Heavy-heavy terms

By (46), $H_+ = \sum_r \mathcal{S}_r(R) \mathcal{S}_r(R^\dagger)^\dagger$. Corollary 6.2 with $Z = W = R$ and Proposition 6.3 give

$$\|H_+|_{\mathcal{K}_\nu}\|^2 \leq \|\mathcal{G}(R^\dagger)|_{\mathcal{K}_\nu}\| \cdot \|\mathcal{G}(R)|_{\mathcal{K}_{\nu+2}}\| \leq \nu(\nu+2), \quad \text{so} \quad \|H_+|_{\mathcal{K}_\nu}\| \leq \nu+1. \quad (70)$$

By (49) and Proposition 6.3, on $\mathcal{K}_\nu$,

$$-\nu I \preceq -D_H \preceq H_0 = \mathcal{G}(R) + \mathcal{G}(R^\dagger) - D_H \preceq 2\nu I, \quad \text{so} \quad \|H_0|_{\mathcal{K}_\nu}\| \leq 2\nu. \quad (71)$$

Since $h/m = (b-1)/(sb) \leq 1/s$,

$$\left\| \frac{h}{m} H_+ \Big|_{\mathcal{K}_\nu} \right\| \leq \frac{\nu+1}{s}, \quad \left\| \frac{h}{m} H_0 \Big|_{\mathcal{K}_\nu} \right\| \leq \frac{2\nu}{s}. \quad (72)$$

6.4 Light-light terms

By (46), $L_+ = \sum_r \mathcal{S}_r(P^\dagger) \mathcal{S}_r(P)^\dagger$. Corollary 6.2 with $Z = W = P^\dagger$ and Proposition 6.3 give

$$\left\| \frac{1}{m} L_+ \Big|_{\mathcal{K}_\nu} \right\| \leq \frac{1}{m} \sqrt{\|\mathcal{G}(P)|_{\mathcal{K}_\nu}\| \cdot \|\widehat{G}|_{\mathcal{K}_{\nu+2}}\|} \leq \frac{1}{m} \sqrt{(m+\nu)(d+\nu+1)}. \quad (73)$$

This is the geometric mean of the input budget $m+\nu$ and the output budget $d+\nu+1$, divided by m . For the grade-preserving term, (48) and Proposition 6.3 give

$$\left\| \frac{1}{m} L_0 \Big|_{\mathcal{K}_\nu} \right\| \leq \frac{\|\widehat{G}|_{\mathcal{K}_\nu}\| + \|D_1|_{\mathcal{K}_\nu}\| + \|H'|_{\mathcal{K}_\nu}\|}{m} \leq \frac{(d+\nu-1) + 2\nu}{m} \leq \frac{3(d+\nu)}{m} \leq \frac{3(d+\nu+1)}{m}. \quad (74)$$

Remark 6.4 (Sharpness at $\nu = 0$). The bound (73) cannot be improved in general. For $x \in \mathbb{R}^d$, the vector $L_+(x \otimes \Omega)$ has, for each row r and each unordered pair $\{i, j\}$, the coefficient $(U_{ij} + U_{ji})x$ on the basis vector with light sites (r, i) and (r, j) , so a direct computation using (1) gives

$$\|L_+(x \otimes \Omega)\|^2 = m \left[(d+1)\|x\|^2 - 2 \sum_{i=1}^n \|u_i\|^2 (u_i^\top x)^2 \right].$$

Hence $\|m^{-1}L_+|_{\mathcal{K}_0}\|^2 \leq (d+1)/m$, with equality in the limit of frames with $\max_i \|u_i\| \rightarrow 0$; this matches (73) at $\nu = 0$.

6.5 Mixed terms

By (47), $X_+ = \sum_r \mathcal{S}_r(P^\dagger) \mathcal{S}_r(R^\dagger)^\dagger$ and $X_0 = \sum_r \mathcal{S}_r(P^\dagger) \mathcal{S}_r(R)^\dagger$. Corollary 6.2 with $(Z, W) = (P^\dagger, R)$ and $(P^\dagger, R^\dagger)$, together with Proposition 6.3, gives

$$\|X_+|_{\mathcal{K}_\nu}\| \leq \sqrt{\nu(d+\nu+1)}, \quad \|X_0|_{\mathcal{K}_\nu}\| \leq \sqrt{\nu(d+\nu-1)} \leq \sqrt{\nu(d+\nu)}. \quad (75)$$

The bounds for Y_+ and Y_0 are those of Corollary 5.2. Since X_0 and Y_0 map $\mathcal{K}_\nu$ into itself, $X_0^\dagger$ and $Y_0^\dagger$ have the same norms on $\mathcal{K}_\nu$. To absorb the coefficient $\sqrt{h}$ we use

$$\frac{\sqrt{h}}{m} \sqrt{\nu(d+\nu+1)} \leq \frac{1}{2} \left(\frac{d+\nu+1}{m} + \frac{h\nu}{m} \right), \quad \frac{h\nu}{m} \leq \frac{\nu}{s}, \quad \frac{\sqrt{h}\nu}{m} \leq \frac{\nu}{s}, \quad (76)$$

which follow from $2\sqrt{xy} \leq x+y$, from $h/m \leq 1/s$, and from $\sqrt{b-1} \leq b = m/s$. Consequently

$$\left\| \frac{\sqrt{h}}{m} (X_+ + Y_+) \right\|_{\mathcal{K}_\nu} \leq \frac{d+\nu+1}{m} + \frac{3\nu}{2s}, \quad (77)$$

$$\left\| \frac{\sqrt{h}}{m} (X_0 + X_0^\dagger + Y_0 + Y_0^\dagger) \right\|_{\mathcal{K}_\nu} \leq \frac{d+\nu}{m} + (1+\sqrt{2})\frac{\nu}{s}. \quad (78)$$

(The first term in (77) could be halved; we keep the stated form for the uniform table below.)

6.6 The uniform band envelope

Set

$$\rho_\nu = \sqrt{\frac{d+\nu+1}{m}}, \quad \gamma_\nu = \frac{\nu+1}{s}. \quad (79)$$

Since $\sqrt{\nu/m} \leq \rho_\nu$, the bound (73) gives

$$\left\| \frac{1}{m} L_+ \right\|_{\mathcal{K}_\nu} \leq \rho_\nu \sqrt{1 + \frac{\nu}{m}} \leq \rho_\nu \left(1 + \sqrt{\frac{\nu}{m}} \right) \leq \rho_\nu + \rho_\nu^2, \quad (80)$$

with no restriction on the relative size of ν and m . The estimates of this section are collected in Table 2.

Proposition 6.5 (Uniform band envelope). *For every $\nu \geq 0$ and $\Delta \in \{1, 0, -1\}$,*

$$\|B^\Delta|_{\mathcal{K}_\nu}\| \leq C_{\text{band}} \left[\sqrt{\frac{d+\nu+1}{m}} + \frac{d+\nu+1}{m} + \frac{\nu+1}{s} \right], \quad C_{\text{band}} = 3 + \sqrt{2}. \quad (81)$$

Proof. By (39), the triangle inequality, and Table 2, $\|B^\pm|_{\mathcal{K}_\nu}\| \leq \rho_\nu + 2\rho_\nu^2 + \frac{5}{2}\gamma_\nu$ and $\|B^0|_{\mathcal{K}_\nu}\| \leq 4\rho_\nu^2 + (3 + \sqrt{2})\gamma_\nu$. Every coefficient is at most $3 + \sqrt{2}$. $\square$

Grade shift	Light–light	Light–heavy	Heavy–heavy	Source
+2	$\rho_\nu + \rho_\nu^2$	$\rho_\nu^2 + \frac{3}{2}\gamma_\nu$	γ_ν	(80), (77), (72)
0	$3\rho_\nu^2$	$\rho_\nu^2 + (1 + \sqrt{2})\gamma_\nu$	$2\gamma_\nu$	(74), (78), (72)
−2	$\rho_\nu + \rho_\nu^2$	$\rho_\nu^2 + \frac{3}{2}\gamma_\nu$	γ_ν	adjoint of the +2 row

Table 2: Upper bounds on $\mathcal{K}_\nu$ for the norms of the three components of B^+ , B^0 , and B^- in (39), including the normalization $1/m$ and the coefficients $\sqrt{h}$ and h . Each entry of the last row vanishes when $\nu < 2$; for $\nu \geq 2$ it is the corresponding +2 entry at grade $\nu - 2$, because $B^-|_{\mathcal{K}_\nu}$ is the adjoint of $B^+|_{\mathcal{K}_{\nu-2}}$, and ρ_ν, γ_ν are nondecreasing in ν .

7 Proof of the main theorem

7.1 The moment bound

Fix an integer $q \geq 1$, let $D_q = d + 2q + 1$, and define

$$\beta_q = C_{\text{band}} \left[\sqrt{\frac{D_q}{m}} + \frac{D_q}{m} + \frac{2q+1}{s} \right]. \quad (82)$$

By Proposition 6.5, $\|B^\Delta|_{\mathcal{K}_\nu}\| \leq \beta_q$ for all Δ and all $0 \leq \nu \leq 2q$.

Theorem 7.1 (Moment bound). *For every integer $q \geq 1$,*

$$\mathbb{E} \text{tr}(E_{\text{iid}}^{2q}) \leq d(3\beta_q)^{2q} \quad \text{and} \quad \mathbb{E} \text{tr}(E_{\text{SS}}^{2q}) \leq d(3c_b^2\beta_q)^{2q}. \quad (83)$$

Proof. Let $v_\alpha = q_\alpha \otimes \Omega \in \mathcal{K}_0$. Inserting $I = \sum_\nu \mathcal{P}_\nu$ between consecutive factors of $\mathcal{M}^{2q}$ in (31) and using $\mathcal{M} = B^+ + B^0 + B^-$ with $B^\Delta \mathcal{K}_\nu \subseteq \mathcal{K}_{\nu+2\Delta}$,

$$\langle v_\alpha, \mathcal{M}^{2q} v_\alpha \rangle = \sum_{(\Delta_1, \dots, \Delta_{2q}) \in \{1, 0, -1\}^{2q}} \langle v_\alpha, B^{\Delta_{2q}} \dots B^{\Delta_1} v_\alpha \rangle,$$

where the word indexed by (Δ_t) visits the grades $\nu_t = 2 \sum_{u \leq t} \Delta_u$ and vanishes unless $\nu_{2q} = 0$ and $\nu_t \geq 0$ for all t . A nonvanishing word has the same number $k \leq q$ of upward and downward steps, so $0 \leq \nu_t \leq 2k \leq 2q$ at all times. For such a word, submultiplicativity gives

$$|\langle v_\alpha, B^{\Delta_{2q}} \dots B^{\Delta_1} v_\alpha \rangle| \leq \prod_{t=1}^{2q} \|B^{\Delta_t}|_{\mathcal{K}_{\nu_{t-1}}}\| \leq \beta_q^{2q}.$$

There are 3^{2q} words; summing over them and over $\alpha \in [d]$ gives the first bound in (83), and Proposition 2.2 gives the second. $\square$

7.2 From moments to probability

For a real symmetric matrix A , $\|A\|^{2q} \leq \text{tr}(A^{2q})$, because the right side is the sum of the $2q$ -th powers of the eigenvalues. Markov's inequality and Theorem 7.1 therefore give

$$\mathbb{P}\{\|E_{\text{SS}}\| > \varepsilon\} \leq \frac{\mathbb{E} \text{tr}(E_{\text{SS}}^{2q})}{\varepsilon^{2q}} \leq d \left(\frac{3c_b^2\beta_q}{\varepsilon} \right)^{2q}. \quad (84)$$

Now let q, s, b , and m be as in (6) and (7). Then

$$m \geq \frac{\Lambda^2 D_q}{\varepsilon^2}, \quad s \geq \frac{\Lambda(2q+1)}{\varepsilon}, \quad (85)$$

so that $\sqrt{D_q/m} \leq \varepsilon/\Lambda$, $(2q+1)/s \leq \varepsilon/\Lambda$, and $D_q/m \leq \varepsilon^2/\Lambda^2 \leq \varepsilon/\Lambda$, using $\varepsilon \leq 1 \leq \Lambda$. Hence

$$\beta_q \leq \frac{3C_{\text{band}} \varepsilon}{\Lambda} = \frac{C_{\text{band}} \varepsilon}{30c_\star^2}, \quad \frac{3c_b^2 \beta_q}{\varepsilon} \leq \frac{C_{\text{band}}}{10} = \frac{3 + \sqrt{2}}{10} < 0.442 < \frac{1}{2}, \quad (86)$$

where the second inequality uses $c_b \leq c_\star$ from Lemma 2.1(iii). By (84),

$$\mathbb{P}\{\|E_{\text{SS}}\| > \varepsilon\} \leq d 4^{-q} \leq \frac{\delta^2}{d} \leq \delta, \quad (87)$$

since $2^{-q} \leq \delta/d$ by the definition of q , and $\delta \leq 1 \leq d$. This proves (8); (9) is the equivalent quadratic-form statement noted in Section 1.

7.3 Integer rounding and explicit constants

Put $\lambda = \Lambda(2q+1)/\varepsilon$; then $\lambda \geq 1$ and $s = \lceil \lambda \rceil \leq \lambda + 1 \leq 2\lambda$. Since $2q+1 \leq 3q$,

$$s \leq \frac{\Lambda(2q+1)}{\varepsilon} + 1 \leq \frac{6\Lambda q}{\varepsilon} < \frac{1356q}{\varepsilon}. \quad (88)$$

Since $b = \lceil M_0/s \rceil$, we have $M_0 \leq m = sb < M_0 + s$, and

$$\frac{s}{M_0} \leq \frac{2\Lambda(2q+1)/\varepsilon}{\Lambda^2 D_q / \varepsilon^2} \leq \frac{2\varepsilon}{\Lambda} < 1, \quad (89)$$

using $2q+1 \leq D_q$. In particular $M_0/s > 1$, so

$$b = \left\lceil \frac{M_0}{s} \right\rceil \geq 2, \quad (90)$$

which is the fact used in Section 3, and $m < M_0 + s < 2M_0$. Using $2M_0 < 3\Lambda^2 D_q / \varepsilon^2$ and $D_q = d + 2q + 1 \leq 2(d+q)$,

$$m < \frac{6\Lambda^2(d+q)}{\varepsilon^2} < \frac{306456(d+q)}{\varepsilon^2}. \quad (91)$$

This proves (10) and completes the proof of Theorem 1.1.

8 Scope and limitations

The proof uses, in order: the coupling and moment transfer of Section 2; the vacuum-moment identity (31) and the band inventory (39); the light-sector inequality, Lemma 4.1; the row-local estimates of Section 5; the block Cauchy–Schwarz bounds of Section 6, summarized in Proposition 6.5; and the moment expansion, Markov inequality, and rounding of Section 7. Only the last step uses the specific parameter choice (7), and only through (85) and (90).

The following restrictions are part of the statement.

- All hash values and signs are independent. The $2q$ -th moment of the Gram error involves up to $4q$ column selectors, so moment matching of order $2q$ at the level of the Gram error does not follow from $2q$ -wise independence of the primitive hashes and signs; a limited-independence version would require a separate joint-moment computation for the chosen encoding.
- The matrix U is fixed, or independent of Π (Remark 1.2); an adaptively chosen U is not covered.

- The result concerns the SparseStack distribution (4), not an arbitrary sparse embedding with s nonzeros per column.
- The constants have not been optimized. The dominant losses are the factor $c_\star^2 < 2.51$ from Proposition 2.2, the factor 3 from the number of bands, and the factor 3 from the three terms of the envelope; together with C_{band} they account for the choice $\Lambda = 90c_\star^2$ in (5).

A Verification status

Version history. Versions through Version 1.6.1 circulated under the title “A Sparse-Fock Proof of the Upper Edge for Fully Independent SparseStack.” The main theorem, parameter choices, final band envelope, and final numerical constants are unchanged; Version 1.8 reorganizes and expands the exposition. Version 2.0 revises the abstract and adds related-work attributions in Sections 1.3, 2, and 4; the mathematical results and proofs are unchanged.

The fixed-matrix theorem of this paper has been formalized in Lean 4 [14]. The artifact uses Lean 4.33.0 and mathlib [23] commit [db584cd6d46c92f209a44c0f1c829460d327499d](#). The statements below concern that artifact: “kernel-checked” means that Lean accepted the stated declarations after elaboration, not that Lean checked the prose of this paper.

Availability. The [project repository](#) contains the canonical Version 1.5 manuscript, the formal source, and the audit record. This correspondence audit used immutable repository commit [b9da2b5d96e5](#) (the link records the full hash). From the repository root, the artifact builds with `cd formal && lake build`.

Main statement. The final matrix-level declaration is `SparseFock.MainTheorem.fully_independent_sparseStack_matrix`. Its hypotheses are $d \geq 1$, $0 < \varepsilon \leq 1$, $0 < \delta \leq 1$, a real matrix U , and $U^\top U = I_d$; its conclusion packages the SparseStack law, the failure bound (8), the two-sided event (9), exact column sparsity, the bounds (10), $b \geq 2$, and $m = sb$. The declaration has no further hypotheses. The source signature is given in [formal/SparseFockFormal/MainTheorem.lean](#). The frame-level endpoint is `SparseFock.MainTheorem.fully_independent_sparseStack`, and the finite-law conditioning corollary for a random frame independent of the sketch (Remark 1.2) is

`SparseFock.MainTheorem.fully_independent_sparseStack_independent_frame`.

Axiom audit. Running `#print axioms` on the main declaration reports only `propext`, `Classical.choice`, and `Quot.sound`. A source sweep confirms that the project contains none of `sorry`, `admit`, project-defined `axiom`, `unsafe`, `partial`, `opaque`, `implemented_by`, and `native_decide`.

Correspondence with the paper. The Lean artifact was developed against Version 1.5. Direct comparison with Theorem 1.1 in Version 2.0 confirms the same real, fully independent signed-hash SparseStack law and normalization; the same definitions of q, D_q, s, M_0, b, m ; the same constants $c_\star, C_{\text{band}}, \Lambda$; and the same operator-failure event, two-sided OSE event, exact column sparsity, strict dimension bounds, $b \geq 2$, and $m = sb$. Version 2.0 states $n \geq 1$ explicitly; in the Lean theorem this follows from $d \geq 1$ and $U^\top U = I_d$.

The proof organization is not line-by-line identical. The three directional uses and two heavy-leg uses of the merged row-local hop Lemma 5.1 are kernel-checked separately in `DirectionalConcrete`, `DirectionalNormalized`, and `HeavyBandsConcrete`. The row-operator notation $\mathcal{S}_r(Z), \mathcal{G}(Z)$ of Section 3.7 packages concrete factorizations checked in

`TypedBlockCS` and the concrete heavy, light, and mixed band modules. The general vacuum-moment identity is formalized; Example 3.2 is explanatory. The final band envelope is checked in `ConcreteBandEnvelope`, and the moment expansion, Markov step, Gram/OSE equivalence, and integer rounding are checked in the endgame modules. Thus the fixed-matrix theorem and every load-bearing concrete estimate used to prove it are kernel-checked, while some generic notation, merged lemmas, examples, and sharper non-load-bearing remarks are expository consolidations in this version.

What is not verified. The kernel check certifies the formal derivation. The theorem-level correspondence reported above is a human comparison, not a judgment made by Lean. Lean does not certify the prose, the explanatory examples, attribution, novelty, or the claim that the formal statement captures the intended external problem; those remain matters for conventional review.

B Notation index

Symbol	Meaning
<i>Model and parameters</i>	
n, d, U, u_i, U_{ij}	Ambient dimension, subspace dimension, fixed $U \in \mathbb{R}^{n \times d}$ with $U^\top U = I_d$, its rows $u_i \in \mathbb{R}^d$, and $U_{ij} = u_i u_j^\top$.
$\varepsilon, \delta, q, D_q, M_0$	Distortion, failure probability, $q = \max\{1, \lceil \log_2(d/\delta) \rceil\}$, $D_q = d + 2q + 1$, $M_0 = \Lambda^2 D_q / \varepsilon^2$.
s, b, m, h	Blocks (nonzeros per column), rows per block, total rows $m = sb$, and $h = b - 1$.
$H_g, h_g(i), \sigma_{gi}, \Pi$	CountSketch blocks, hash rows, signs, and the SparseStack matrix (4).
$c_\star, C_{\text{band}}, \Lambda$	The constants (5).
$\xi_{gi}, \zeta_{gi}, p_0, c_b$	Signed one-hot selector, its independent-entry proxy, $p_0 = \mathbb{P}\{\zeta = 0\}$, and $c_b = (1 - p_0)^{-1}$.
η_{ri}	Three-point entries (18) indexed by sites.
$E_{\text{SS}}, E_{\text{iid}}$	Hollow Gram errors (13) and (19).
<i>Tensor-product model</i>	
$\mathbf{l}, (r, i)$	Site set $[m] \times [n]$ and a site.
$\mathcal{F}, \Omega, e_0, e_1, e_2$	Product space $(\mathbb{R}^3)^{\otimes mn}$, vacuum, and the local basis (20).
$J, P, P^\dagger, R, R^\dagger$	Jacobi matrix (21) and its four legs (22); Z_{ri} is the lift of Z to site (r, i) .
$\tau, \tau\rangle, \text{gr}(\tau)$	Pattern, basis vector, and grade (32).
$S_j(\tau), S_j^r(\tau), F_r(\tau)$	Level- j sites, level- j columns of row r , free columns of row r .
$\tau[a \mapsto j]$	Pattern with site (r, a) reassigned to level j (row r fixed by context).
$\nu_r(\tau), \mathcal{N}_r$	Grade of row r and the corresponding operator.
$\mathcal{H}_\nu, \mathcal{K}_\nu, \pi_\nu, \mathcal{P}_\nu$	Grade- ν sector of $\mathcal{F}$, $\mathcal{K}_\nu = \mathbb{R}^d \otimes \mathcal{H}_\nu$, and the orthogonal projections onto them.
$\mathcal{M}, B^+, B^0, B^-$	Operator (30) and its bands (34).
$\mathbf{K}_{T,\ell}, S; T\rangle, F_T$	Block with heavy set T and ℓ light sites, its basis vectors, and $\mathbf{l} \setminus T$.
<i>Operators</i>	
$L_\bullet, H_\bullet, X_\bullet, Y_\bullet$	Families (36)–(37) in the inventory (39).
$S_r(Z), \mathcal{G}(Z)$	Row operator and Gram operator (40)–(41).
$\widehat{G}, H'$	$\mathcal{G}(P^\dagger)$ and the transpose light hop (45).
D_0, D_1, D_H	Diagonal operators (44).
$\rho_\nu, \gamma_\nu, \beta_q$	Envelope scales (79) and the ladder constant (82).

References

- [1] Luigi Accardi and Marek Bożejko. Interacting Fock spaces and Gaussianization of probability measures. *Infinite Dimensional Analysis, Quantum Probability and Related Topics*, 1(4):663–670, 1998. [doi:10.1142/S0219025798000363](https://doi.org/10.1142/S0219025798000363).
- [2] Noah Amsel et al. Linear systems and eigenvalue problems: Open questions from a Simons workshop. arXiv preprint arXiv:2602.05394, 2026. [doi:10.48550/arXiv.2602.05394](https://doi.org/10.48550/arXiv.2602.05394).
- [3] Afonso S. Bandeira, March T. Boedihardjo, and Ramon van Handel. Matrix concentration inequalities and free probability. *Inventiones Mathematicae*, 234:419–487, 2023. [doi:10.1007/s00222-023-01204-6](https://doi.org/10.1007/s00222-023-01204-6).

- [4] Afonso S. Bandeira, Giorgio Cipolloni, Dominik Schröder, and Ramon van Handel. Matrix concentration inequalities and free probability II. Two-sided bounds and applications. arXiv preprint arXiv:2406.11453, 2024; revised 2025. doi:10.48550/arXiv.2406.11453.
- [5] Tatiana Brailovskaya and Ramon van Handel. Universality and sharp matrix concentration inequalities. *Geometric and Functional Analysis*, 34(6):1734–1838, 2024. doi:10.1007/s00039-024-00692-9.
- [6] Artur Buchholz. Operator Khintchine inequality in non-commutative probability. *Mathematische Annalen*, 319(1):1–16, 2001. doi:10.1007/PL00004425.
- [7] Chris Camaño, Ethan N. Epperly, Raphael A. Meyer, and Joel A. Tropp. Faster linear algebra algorithms with structured random matrices. arXiv preprint arXiv:2508.21189, 2025. doi:10.48550/arXiv.2508.21189.
- [8] Kenneth L. Clarkson and David P. Woodruff. Low rank approximation and regression in input sparsity time. In *Proceedings of the 45th Annual ACM Symposium on Theory of Computing (STOC 2013)*, pages 81–90. ACM, 2013. doi:10.1145/2488608.2488620.
- [9] Moses Charikar, Kevin C. Chen, and Martin Farach-Colton. Finding frequent items in data streams. In *Automata, Languages and Programming (ICALP 2002)*, volume 2380 of *Lecture Notes in Computer Science*, pages 693–703. Springer, 2002. doi:10.1007/3-540-45465-9_59.
- [10] Shabarish Chenakkod, Michał Dereziński, Xiaoyu Dong, and Mark Rudelson. Optimal embedding dimension for sparse subspace embeddings. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC 2024)*, pages 1106–1117. ACM, 2024. doi:10.1145/3618260.3649762.
- [11] Shabarish Chenakkod, Michał Dereziński, and Xiaoyu Dong. Optimal oblivious subspace embeddings with near-optimal sparsity. In *52nd International Colloquium on Automata, Languages, and Programming (ICALP 2025)*, volume 334 of *LIPICs*, article 55, pages 55:1–55:20. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2025. doi:10.4230/LIPICs.ICALP.2025.55.
- [12] Shabarish Chenakkod, Michał Dereziński, and Xiaoyu Dong. Optimal subspace embeddings: Resolving Nelson–Nguyen conjecture up to sub-polylogarithmic factors. In *Proceedings of the 2026 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 4501–4559. SIAM, 2026. doi:10.1137/1.9781611978971.163.
- [13] Michael B. Cohen. Nearly tight oblivious subspace embeddings by trace inequalities. In *Proceedings of the Twenty-Seventh Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 278–287. SIAM, 2016. doi:10.1137/1.9781611974331.ch21.
- [14] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021. doi:10.1007/978-3-030-79876-5_37.
- [15] David Gross and Vincent Nesme. Note on sampling without replacing from a finite collection of matrices. arXiv preprint arXiv:1001.2738, 2010. doi:10.48550/arXiv.1001.2738.
- [16] Uffe Haagerup and Gilles Pisier. Bounded linear operators between C^* -algebras. *Duke Mathematical Journal*, 71(3):889–925, 1993. doi:10.1215/S0012-7094-93-07134-7.

- [17] Yukihiro Hashimoto, Akihito Hora, and Nobuaki Obata. Central limit theorems for large graphs: Method of quantum decomposition. *Journal of Mathematical Physics*, 44(1):71–88, 2003. doi:[10.1063/1.1528275](https://doi.org/10.1063/1.1528275).
- [18] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi:[10.1080/01621459.1963.10500830](https://doi.org/10.1080/01621459.1963.10500830).
- [19] Akihito Hora and Nobuaki Obata. *Quantum Probability and Spectral Analysis of Graphs*. Theoretical and Mathematical Physics. Springer, Berlin, 2007. doi:[10.1007/3-540-48863-4](https://doi.org/10.1007/3-540-48863-4).
- [20] Han Huang, Mark Rudelson, and Konstantin Tikhomirov. Well-invertible column subsets of sparse matrices are rare. arXiv preprint arXiv:2607.05384, 2026. doi:[10.48550/arXiv.2607.05384](https://doi.org/10.48550/arXiv.2607.05384).
- [21] Daniel M. Kane and Jelani Nelson. Sparser Johnson–Lindenstrauss transforms. *Journal of the ACM*, 61(1):Article 4, 4:1–4:23, 2014. doi:[10.1145/2559902](https://doi.org/10.1145/2559902).
- [22] Alastair Kay, Ravishankar Ramanathan, and Dagomir Kaszlikowski. Optimal asymmetric quantum cloning for quantum information and computation. *Quantum Information and Computation*, 13(9–10):880–900, 2013. doi:[10.26421/QIC13.9-10-9](https://doi.org/10.26421/QIC13.9-10-9). arXiv:[1208.5574](https://arxiv.org/abs/1208.5574).
- [23] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020. doi:[10.1145/3372885.3373824](https://doi.org/10.1145/3372885.3373824).
- [24] Jelani Nelson and Huy L. Nguyen. OSNAP: Faster numerical linear algebra algorithms via sparser subspace embeddings. In *2013 IEEE 54th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 117–126. IEEE, 2013. doi:[10.1109/FOCS.2013.21](https://doi.org/10.1109/FOCS.2013.21).
- [25] Jelani Nelson and Huy L. Nguyen. Lower bounds for oblivious subspace embeddings. In *Automata, Languages, and Programming (ICALP 2014)*, volume 8572 of *Lecture Notes in Computer Science*, pages 883–894. Springer, 2014. doi:[10.1007/978-3-662-43948-7_73](https://doi.org/10.1007/978-3-662-43948-7_73).
- [26] Goutham Rajendran and Madhur Tulsiani. Concentration of polynomial random matrices via Efron–Stein inequalities. In *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 3614–3653. SIAM, 2023. doi:[10.1137/1.9781611977554.ch138](https://doi.org/10.1137/1.9781611977554.ch138).
- [27] Moshe Shaked and J. George Shanthikumar. *Stochastic Orders*. Springer Series in Statistics. Springer, New York, 2007. doi:[10.1007/978-0-387-34675-5](https://doi.org/10.1007/978-0-387-34675-5).
- [28] Joel A. Tropp. Improved analysis of the subsampled randomized Hadamard transform. *Advances in Adaptive Data Analysis*, 3(1–2):115–126, 2011. doi:[10.1142/S1793536911000787](https://doi.org/10.1142/S1793536911000787).
- [29] Joel A. Tropp. Comparison theorems for the extreme eigenvalues of a random symmetric matrix. arXiv preprint arXiv:2603.04365v4, 2026. doi:[10.48550/arXiv.2603.04365](https://doi.org/10.48550/arXiv.2603.04365).